EIG-MPHBS: a rotation-invariant information exchange for hybrid Honey Badger–Marine Predators optimisation

Abstract

Hybrid metaheuristics that run two populations side by side often share information by recombining their members coordinate by coordinate. This exchange depends on the coordinate system: when the variables of a problem are coupled by a rotation, recombining along the original axes breaks the structure that each population has found. We study this weakness in MPHBS, a recent hybrid of the Honey Badger Algorithm and the Marine Predators Algorithm, and propose EIG-MPHBS, which performs the exchange in the eigenvector frame of the covariance of both populations in half of the iterations. The exchange step becomes provably invariant to rotations of the search space, uses exactly the same number of function evaluations as the original and adds 3.2–5.5% to the run time. We evaluate EIG-MPHBS in 18,840 runs on CEC2022 ($D = 10, 20$), the 24 BBOB functions ($D = 20$) and eight FIR filter-design problems, against MPHBS, its parent algorithms, four classical metaheuristics, L-SHADE and CMA-ES. In a test fixed before the experiments on 28 held-out instances, EIG-MPHBS improves on both MPHBS and its budget-matched random-mediator variant (Holm-adjusted $p = 0.043$); the margin is small and depends on how ties are treated. On all 56 instances it is never significantly worse than MPHBS, reduces the median error on the rotated ellipsoid 14.3-fold, and beats HBA, MPA, PSO, DE, GWO and WOA on 260 of 336 comparisons. It ranks third of eleven overall, behind L-SHADE and CMA-ES, and remains behind the CEC2022 entry EA4eig.

Keywords: Hybrid metaheuristics, Rotation invariance, Eigenvector crossover, Honey Badger Algorithm, Marine Predators Algorithm, FIR filter design

1. Introduction

Many decision problems handled by expert and intelligent systems end in a continuous optimisation problem whose objective is available only as a black box: the parameters of a controller evaluated by simulation, the coefficients of a digital filter evaluated through its frequency response, the hyper-parameters of a model evaluated by cross-validation, or the settings of an energy system evaluated by a load-flow solver. Gradient information is unavailable or unreliable in these settings, the objective is often multimodal, and each evaluation can be expensive. Population-based metaheuristics are a common choice for such problems because they need only objective values, are easy to embed in an application pipeline, and can be run in parallel.

No single metaheuristic performs best on all problems (Wolpert and Macready, 1997), and this has made hybridisation one of the most popular design strategies in the field (Talbi, 2002; Blum et al., 2011). A hybrid combines algorithms with complementary search behaviour,

typically an explorative one and an exploitative one, and lets them share information. Multi-population hybrids run the algorithms on separate populations and exchange information between them at regular intervals (Alba and Tomassini, 2002; Ma et al., 2019). How this exchange is designed largely determines whether the hybrid is better than its components: an exchange that moves too much information makes the populations collapse onto each other, and one that moves too little leaves them independent.

MPHBS (Flores and Olivares, 2026) is a recent multi-population hybrid of the Honey Badger Algorithm (HBA) (Hashim et al., 2022) and the Marine Predators Algorithm (MPA) (Faramarzi et al., 2020), two widely used metaheuristics. In every iteration, after both algorithms have moved their own population, a mediator assembles new candidate solutions dimension by dimension from the best members of both populations: coordinate d of a candidate is copied from a chosen member of the HBA or the MPA population. MPHBS was proposed with a SARSA-style reinforcement learner that decides which population and which rank supply each coordinate. In our experiments a uniform random choice at the same evaluation budget performs as well as this learner (mean ranks 4.62 and 4.64, Section 7.4), which points to the exchange step itself, rather than the learning, as the component that matters.

This paper starts from that observation and asks a different question: what limits the exchange? Our answer is that it depends on the coordinate system. Copying coordinate d from one good solution and coordinate d' from another is a sound way to combine partial solutions when the variables are separable. When the objective couples variables, for instance through a rotation of the search space, the meaningful directions are not the coordinate axes, and a candidate assembled from the coordinates of different solutions lies far from all of them along those directions. This weakness of coordinate-wise recombination is well known for genetic algorithms (Salomon, 1996) and differential evolution (Guo and Yang, 2015), and it is visible in MPHBS: on the BBOB rotated ellipsoid (f10), MPHBS ends with a median error of 121, while on the separable ellipsoid (f2) it reaches the 10^{-8} accuracy floor (Section 7.2).

The remedy we adopt is known in other algorithm families. Eigenvector-based crossover in differential evolution recombines vectors in the eigenvector frame of the population covariance (Guo and Yang, 2015; Wang et al., 2014), EA4eig uses eigen crossover inside a cooperative model of four algorithms (Bujok and Kolenovský, 2022), and in evolutionary multitasking knowledge is transferred between populations in aligned eigen-subspaces (Tang et al., 2021, 2022). We transfer this principle to the dimension-wise exchange of HBA–MPA hybrids and propose EIG-MPHBS, in which the exchange is carried out in the eigenvector frame of the pooled covariance of both populations in half of the iterations and along the original axes in the other half. We do not claim a new operator. The contribution lies in identifying the coordinate dependence of the exchange as a cause of the hybrid's weakness, fixing it at no extra evaluation cost, explaining why the fix works, and testing the result rigorously against a broad set of algorithms.

The study is organised around four research questions:

RQ1 Does the eigen-frame exchange improve MPHBS on instances that were not used to design it?

RQ2 Does it remove the dependence of the exchange on rotations of the search space, and by what mechanism?

RQ3 How does EIG-MPHBS compare with the parent algorithms, with widely used metaheuristics and with state-of-the-art optimisers?

RQ4 What does the change cost, and how sensitive is it to its single parameter?

The main contributions are:

1. EIG-MPHBS, a hybrid HBA–MPA optimiser whose inter-population exchange operates in the eigenvector frame of the pooled populations with probability $q = 0.5$ (Section 4, Fig. 1).
2. Two properties: the eigen-frame exchange is invariant to rotations of the search space, and it uses exactly the same number of function evaluations as the axis-frame exchange, at an extra arithmetic cost of $O(ND^2 + D^3)$ per iteration (Section 5).
3. An evaluation on 56 instances from CEC2022, BBOB and an FIR filter-design application against ten algorithms: MPHBS and its random-mediator variant, the parent algorithms HBA and MPA, four widely used metaheuristics (PSO, DE, GWO, WOA), and two state-of-the-art references, L-SHADE and CMA-ES, plus a comparison with the official CEC2022 results of EA4eig (Section 7).
4. A mechanism analysis showing that the eigen frame raises the acceptance rate of exchanged candidates on coupled problems, an ablation and sensitivity study, and an analysis of the designed filters (Sections 7.3 to 7.7).
5. A transparent protocol: the main hypotheses, the held-out instances and the statistical tests were fixed in a design document before any confirmatory run, and code, raw results and the scripts that produce every number in this paper are public.

The results are positive but bounded. EIG-MPHBS improves on MPHBS, is never significantly worse than it on any of the 56 instances, and clearly outperforms the common metaheuristics, including both of its parent algorithms. L-SHADE, CMA-ES and EA4eig remain stronger on most instances, and the confirmatory advantage on the held-out instances, while significant under the test we fixed in advance, is small. We report both sides.

The rest of the paper is organised as follows. Section 2 reviews related work, Section 3 describes MPHBS in detail, Section 4 presents EIG-MPHBS and Section 5 its properties. Section 6 describes the experimental setup and Section 7 the results. Section 8 discusses them and Section 9 concludes.

2. Related work

2.1. HBA, MPA and their hybrids

HBA (Hashim et al., 2022) models the foraging of the honey badger. Each agent either "digs", moving around the best agent with a step scaled by a smell intensity that grows with the distance to the prey, or follows a honeyguide towards the best agent; a density factor that decreases over the run controls the step sizes. A recent survey counts 52 studies proposing improved HBA variants, for example with chaotic maps, Lévy flights or opposition-based learning, 20 studies that hybridise it with other methods, and over a hundred applications (Hassan et al., 2024). MPA (Faramarzi et al., 2020) models the foraging of marine predators. It divides the run into three phases with different velocity ratios between predator and prey, uses

Brownian motion in the first phase, a mixture of Brownian and Lévy motion in the second and Lévy motion in the third, and adds two mechanisms: a marine memory that keeps the better of the old and the new position of each agent, and a fish-aggregating-device (FAD) perturbation that occasionally makes long jumps. A review of 102 MPA studies discusses its variants, hybrids and limitations, among them premature convergence on some problem classes (Al-Betar et al., 2023).

HBA and MPA have complementary behaviour, which makes them natural candidates for hybridisation. Al-Dhaifallah et al. combined them in series, population-division and switching schemes for an electric-vehicle hosting-capacity problem (Al-Dhaifallah et al., 2024). MPHBS (Flores and Olivares, 2026) runs both on separate populations and couples them through a mediator that recombines their members dimension by dimension under the control of a SARSA learner.

2.2. Multi-population hybrids and information exchange

Talbi's taxonomy distinguishes low-level hybrids, which replace a component of one algorithm with another, from high-level hybrids, in which self-contained algorithms cooperate, and relay hybrids, which run them in sequence, from teamwork hybrids, which run them in parallel (Talbi, 2002; Blum et al., 2011). Island and multi-population models have long studied the design of migration: what is exchanged, between which populations, how often and how much (Alba and Tomassini, 2002; Ma et al., 2019). Classical migration moves whole individuals. The MPHBS mediator instead builds new individuals from coordinates of members of both populations, so it acts like a multi-parent crossover between populations and inherits the properties, and the weaknesses, of coordinate-wise crossover. Population diversity, which the exchange directly affects, is a central quantity for understanding the balance between exploration and exploitation (Črepinšek et al., 2013; Squillero and Tonda, 2016; Morales-Castañeda et al., 2020).

2.3. Learning-based control of the exchange

Machine learning is increasingly used to control metaheuristics (Talbi, 2021). Parameter control (Karafotias et al., 2015) and adaptive operator selection with bandits (Fialho et al., 2010) were followed by reinforcement-learning controllers (Drugan, 2019; Song et al., 2024), for example Q-learning to switch between exploration and exploitation in grey wolf and whale optimisers (Seyyedabbasi et al., 2021) or to set the parameters of the sine cosine algorithm (Hamad et al., 2022). MPHBS belongs to this line. EIG-MPHBS takes a different route: it uses a simple random mediator and changes the space in which the mediator works rather than the way it decides.

2.4. Rotation invariance

An algorithm is rotation invariant if its behaviour on $f(xQ)$ for an orthogonal Q is the rotated image of its behaviour on $f(x)$. Salomon (1996) showed that genetic algorithms with coordinate-wise operators lose much of their performance when a separable function is rotated, because their success relies on the independence of the variables. The same applies to binomial crossover in DE and to any operator that mixes solutions coordinate by coordinate. CMA-ES adapts a full covariance matrix and is rotation invariant by design (Hansen and

Ostermeier, 2001), which is one reason for its strength on ill-conditioned problems. Benchmark suites such as BBOB (Hansen et al., 2009, 2021) and CEC2022 (Kumar et al., 2021) therefore include rotated functions, and BBOB contains separable and rotated versions of the same functions, which we use to measure rotation dependence directly.

2.5. Eigenvector-based recombination

Table 1 summarises the operators that recombine solutions in a coordinate system derived from the population. Guo and Yang (2015) perform DE crossover in the eigenvector frame of the population covariance with a given probability, so that crossover follows the principal directions of the population; CoBiDE (Wang et al., 2014) uses covariance matrix learning in the same way, together with a bimodal parameter distribution. Civicioglu and Besdok (2018) make crossover rotation invariant with a QR decomposition. EA4eig (Bujok and Kolenovský, 2022) lets CMA-ES, CoBiDE, jSO and IDE cooperate and applies eigen crossover in each; it was among the strongest entries of the CEC2022 competition. In evolutionary multitasking, Tang et al. (2021) project the individuals of two task populations onto aligned eigen-subspaces before recombining them, and Tang et al. (2022) transfer knowledge between tasks in a coordinate system built from the major eigenvectors of both populations. QLRUN (Adam et al., 2027) combines Q-learning with covariance matrix learning in a single metaheuristic.

Table 1: Recombination in population-derived coordinate systems (from the cited papers' descriptions).

Work	Host algorithm	Populations	Use of the eigen / rotated frame
Guo and Yang (2015)	DE	one	crossover in the eigenvector frame of the population, with a probability
CoBiDE (Wang et al., 2014)	DE	one	crossover in the frame of the learned covariance
A+ (Civicioglu and Besdok, 2018)	evolutionary search	one	QR-based rotation-invariant crossover
EA4eig (Bujok and Kolenovský, 2022)	CMA-ES, CoBiDE, jSO, IDE	four cooperating	eigen crossover inside each algorithm
Tang et al. (2021)	multitask EA	one per task	inter-task transfer in aligned eigen-subspaces
Tang et al. (2022)	multifactorial EA	one per task	inter-task transfer in a frame from both populations' eigenvectors
QLRUN (Adam et al., 2027)	RUN	one	covariance matrix learning plus Q-learning
EIG-MPHBS (this work)	HBA + MPA hybrid	two heterogeneous	dimension-wise inter-population exchange in the frame of the pooled covariance, with probability q

2.6. Research gap

Exchanging information between populations in an eigen frame of their joint distribution is therefore known, and the closest precedents are the inter-task transfers of Tang et al. (2021, 2022). A Scopus search (title, abstract and keywords, 28 October 2026; queries in the repository) found no study that applies it to the dimension-wise exchange of HBA–MPA

hybrids, or that analyses how such mediators depend on the coordinate system. This matters for practitioners, because hybrids of recent metaheuristics are widely used in applications, often with mediators of this kind, and their weakness on coupled variables is rarely examined. Our contribution is an application of a known principle to this class of hybrids, with a proof of its invariance in this setting, an analysis of its mechanism and cost, and a comparison against a wide set of algorithms under a design fixed in advance.

3. The MPHBS hybrid

3.1. Notation

We minimise $f : \Omega \to \mathbb{R}$ over a box $\Omega = [\ell, u]^D$; points are row vectors. MPHBS maintains an HBA population $X^H \in \mathbb{R}^{N\times D}$ and an MPA population $X^M \in \mathbb{R}^{N\times D}$ with fitness values f_i^H, f_i^M, the best members p^H, p^M of each population, and the global best $p^\star$ with value $f^\star$. MPA keeps a marine memory $(\tilde{X}^M, \tilde{f}^M)$. $t = 0, \dots, T-1$ is the iteration, clip the projection onto Ω, $\odot$ the element-wise product and $U(a, b)$ a uniform random number. An iteration consists of the native moves (Phase 1), an optional FAD stage, and the exchange (Phase 2).

3.2. Phase 1: native moves

HBA. With the smell intensity $I_i = r_i \, \|x_i^H - x_{i+1}^H\|^2/(4\pi\|x_i^H - p^H\|^2 + \epsilon)$, the density factor $\alpha_t = C\exp(-(t+1)/T)$, a random sign $F \in \{-1, 1\}$ and $d_{ij} = p_j^H - x_{ij}^H$, agent i proposes

$$\hat{x}_{ij} = \begin{cases} p_j^H + F\beta I_i \, p_j^H + F r_3 \alpha_t d_{ij} \, \psi_{ij} & r < 0.5 \quad \text{(digging)}, \\ p_j^H + F r_7 \alpha_t d_{ij} & \text{otherwise} \quad \text{(honey)}, \end{cases} \tag{1}$$

with $\psi_{ij} = \Big|\cos(2\pi r_4)\Big(1 - \cos(2\pi r_5)\Big)\Big|$, $\beta = 6$, $C = 2$ and $r, r_3, \dots, r_7 \sim U(0, 1)$. The proposal replaces x_i^H if it is better.

MPA. With the elite $E = p^M$, $P = 0.5$, the adaptive factor $\mathrm{CF}_t = (1 - t/T)^{2t/T}$, Brownian steps $R_B \sim \mathcal{N}(0, 1)^{N\times D}$ and Lévy steps $R_L = 0.05\,\mathrm{Lévy}(1.5)$, agent i moves by

$$x_i^M \leftarrow \begin{cases} x_i^M + PR \odot R_B \odot (E - R_B \odot x_i^M) & \text{phase I}, \\ E + P\,\mathrm{CF}_t \, R_B \odot (R_B \odot E - x_i^M) & \text{phase II}, \ i > \frac{N}{2}, \\ x_i^M + PR \odot R_L \odot (E - R_L \odot x_i^M) & \text{phase II}, \ i \le \frac{N}{2}, \\ E + P\,\mathrm{CF}_t \, R_L \odot (R_L \odot E - x_i^M) & \text{phase III}, \end{cases} \tag{2}$$

where phases I, II, III are $t < T/3$, $T/3 \le t < 2T/3$ and $t \ge 2T/3$, and $R \sim U(0, 1)^D$. After evaluation, the marine memory keeps the old position where it was better and is then updated:

$$x_i^M \leftarrow \begin{cases} \tilde{x}_i^M & \tilde{f}_i^M < f(x_i^M), \\ x_i^M & \text{otherwise}, \end{cases} \qquad (\tilde{X}^M, \tilde{f}^M) \leftarrow (X^M, f^M). \tag{3}$$

Phase 1 costs $2N$ function evaluations (FEs).

FAD stage. With probability FADs = 0.2 the MPA population is perturbed by $\mathrm{CF}_t\,(\ell + R \odot (u-\ell)) \odot B$ with a Bernoulli(0.2) mask B, otherwise by $\Big(\mathrm{FADs}(1-r)+r\Big)(x^M_{\pi_1(i)} - x^M_{\pi_2(i)})$ for random permutations π_1, π_2. Every candidate is evaluated (N FEs) and accepted greedily. The stage is placed before Phase 2 on CEC2022 and BBOB and after it on FIR, as in the original design of MPHBS.

3.3. Phase 2: the dimension-wise exchange

Ranked subsets. From each population the mediator draws a subset of

$$N_{\mathrm{sub}} = \min\Big(N, \max(5, \mathrm{round}(\rho N))\Big) \tag{4}$$

members, half of them the best and the rest at random, and sorts it by fitness. The subsets form a tensor $R \in \mathbb{R}^{N_{\mathrm{sub}} \times D \times 2}$, where $R_{n,d,c}$ is coordinate d of the rank-n member of source $c \in \{H, M\}$.

Episodes. The mediator starts from the current vector $v = p^\star$ (one FE) and runs $E = N_{\mathrm{sub}} N_i$ episodes. The dimensions are split into five blocks (one if $D < 10$) of size $b = \lceil D/5 \rceil$, visited along random permutations. In episode e with active block A_e, K candidates are assembled: candidate k copies v and, for each $d \in A_e$, sets $v'_d = R_{n_d,d,c_d}$ for a source c_d and a rank n_d. In the published MPHBS these choices are made by five policies over a SARSA value table; in the random-mediator variant MPHBS-R, and in EIG-MPHBS, they are drawn uniformly. A proxy score that costs no evaluation,

$$s(v') = -w_{\mathrm{rank}}\,\overline{(n_d+1)}_{d\in A_e} + w_{\mathrm{nov}}\,\overline{|v'_d - v_d|}_{d\in A_e} - w_{\mathrm{best}}\,\|v'_{A_e} - p^\star_{A_e}\|_2, \tag{5}$$

selects one candidate, which favours coordinates from well-ranked members, rewards novelty and penalises distance from the global best. Only the selected candidate is evaluated (one FE per episode).

Acceptance and update. An improving candidate is accepted; a worse one is accepted with probability

$$p_e = \min\Big(0.25,\ 0.2\,(1-g_t)^2\Big(0.35 + 0.65(1-\lambda_e)\Big)\exp\Big(-4\max(0, \Delta_e)\Big)\Big), \tag{6}$$

where $g_t = t/(T-1)$, $\lambda_e = (e-1)/(E-1)$ and $\Delta_e = (f(v') - f(v))/(|f(v)| + \epsilon)$. An accepted candidate replaces the worst member of each subset that it improves; a new global best is recorded and copied into a random row of both subsets. At the end of Phase 2 the subsets are written back into the populations and the best members are synchronised. One iteration costs $3N + 1 + E$ FEs.

3.4. Parameter settings

We keep the parameter values recommended for MPHBS after the parameter study in its original publication (Flores and Olivares, 2026) and do not retune them, so that all three hybrids share the same settings and any difference is due to the exchange alone. With $N = 30$ agents per population: on CEC2022 and BBOB, $K = 7$ candidates per episode, $N_i = 3$ episodes per subset member and $\rho = 0.25$, so $N_{\mathrm{sub}} = 8$ and $E = 24$ episodes per iteration, with the FAD stage before Phase 2; on FIR, $K = 3$, $N_i = 9$ and $E = 72$, with the FAD stage after Phase 2. In all cases $w_{\mathrm{rank}} = 1$, $w_{\mathrm{nov}} = 0.3$ and $w_{\mathrm{best}} = 0.05$.

4. EIG-MPHBS

4.1. Why the exchange depends on the coordinate system

Consider a separable objective $f(x) = \sum_d g_d(x_d)$. If two good solutions a and b are each good in different coordinates, the candidate that takes the good coordinates of each is better than both: coordinate-wise exchange combines building blocks. Now rotate the problem, $f_Q(x) = f(xQ^\top)$ with an orthogonal Q. The building blocks are now the coordinates of $xQ^\top$, i.e. the projections onto the columns of Q, and copying raw coordinates mixes them. For an ill-conditioned rotated function such as the ellipsoid, even two nearby good solutions can produce, by coordinate-wise exchange, a candidate that leaves the narrow valley in which both lie. Such candidates are rejected, the exchange stalls, and the hybrid loses the main channel through which its two populations cooperate.

The populations themselves carry the information needed to correct this. Late in the search, the members of both populations are spread along the valley they occupy, so the principal axes of their covariance approximate the directions in which the objective varies slowly or quickly. Recombining along these axes keeps candidates inside the valley.

4.2. The eigen-frame exchange

At the start of Phase 2, with probability q, EIG-MPHBS computes the sample covariance of the pooled populations and its eigendecomposition,

$$C = \operatorname{cov}\begin{pmatrix} X^H \\ X^M \end{pmatrix} = B\Lambda B^\top, \tag{7}$$

with an orthonormal eigenvector matrix B whose columns are ordered by eigenvalue. Pooling both populations gives $2N$ points for a $D \times D$ estimate and describes the region that the hybrid as a whole is exploring. All operations of the mediator in this iteration are then performed on the eigen coordinates $z = xB$ instead of x:

1. the tensor R holds the projected subsets $S^H B$ and $S^M B$, and the start point is $z = p^\star B$;
2. candidates are assembled coordinate-wise in z, i.e. along the principal axes of the populations;
3. the proxy score (5) is computed on eigen coordinates;
4. the selected candidate is mapped back and clipped to the box, $x' = \mathrm{clip}(z'B^\top)$, and evaluated;
5. if it is accepted, its eigen coordinates are recomputed from the clipped point, $z \leftarrow x'B$, so that the current vector always corresponds to a feasible evaluated point.

With probability $1 - q$ the iteration uses the original axes ($B = I$), and the method is then identical to MPHBS-R. Fig. 1 shows the architecture of the method and Algorithm 1 one iteration.

4.3. Design choices

Random mediator. EIG-MPHBS uses the uniform random mediator of MPHBS-R rather than the SARSA learner, for two reasons. The learner brings no measurable benefit at the same budget (MPHBS and MPHBS-R have nearly identical mean ranks in Section 7.4), and its value table is indexed by coordinates, whose meaning would change whenever the frame changes from one iteration to the next.

Frame probability. We set $q = 0.5$. The axis frame remains useful on separable or nearly separable structure, where coordinate-wise exchange combines building blocks, and the FIR problems show such structure in part. Mixing the two frames keeps both kinds of exchange. The value $q = 0.5$ was chosen over $q = 1$ in a 10-run pilot on 32 instances (BBOB f1, f3, f8, f10, f12, f15, f16, f21; CEC2022 $D = 20$; FIR), reported in Section 7.8; those instances are excluded from the confirmatory test (Section 6.3).

Feasibility. Rotation can move a recombined point outside the box. We clip after mapping back, which keeps the evaluation budget unchanged; the re-projection of accepted points keeps the internal state consistent.

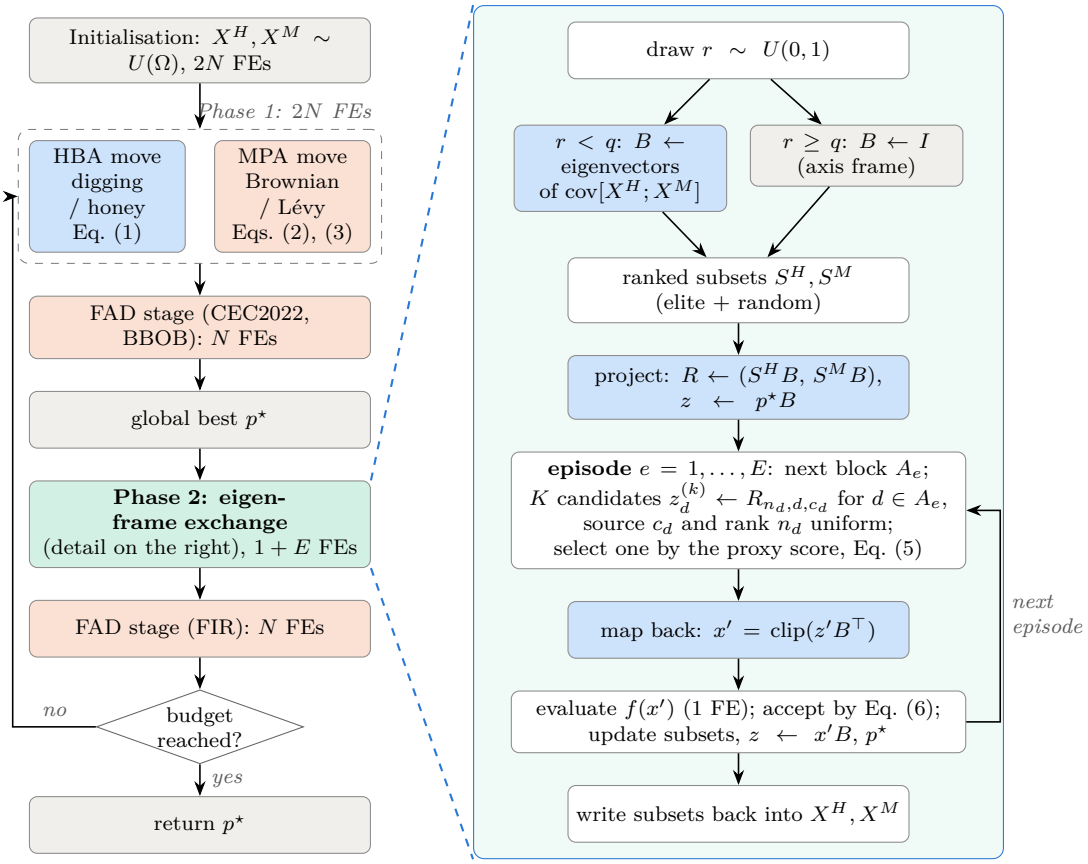


Figure 1: Architecture of EIG-MPHBS. Left: one iteration of the hybrid. Right: the eigen-frame exchange (Phase 2). Blue boxes are the only additions to the random-mediator hybrid MPHBS-R; they use no function evaluations.

Algorithm 1 One iteration of EIG-MPHBS. Lines 4, 5 and 9 differ from MPHBS-R.

Input: populations X^H, X^M, global best $p^\star$, iteration t, eigen-frame probability q

1: HBA move (1) and MPA move (2); evaluate; greedy replacement; marine memory (3) ▷ $2N$ FEs
2: **if** FAD before Phase 2 **then** FAD stage ▷ N
3: $p^\star \leftarrow$ best of p^H, p^M
4: **if** $U(0,1) < q$ **then** $B \leftarrow$ eigenvectors of $\mathrm{cov}([X^H; X^M])$ **else** $B \leftarrow I$ ▷ 0
5: draw ranked subsets S^H, S^M; $R \leftarrow (S^H B, S^M B)$; $z \leftarrow p^\star B$; $f_z \leftarrow f(p^\star)$ ▷ 1
6: **for** $e = 1, \ldots, E$ **do**
7: $\quad A_e \leftarrow$ next block of dimensions
8: $\quad$ for $k = 1..K$: $z^{(k)} \leftarrow z$; for $d \in A_e$ draw (c_d, n_d) uniformly; $z_d^{(k)} \leftarrow R_{n_d, d, c_d}$
9: $\quad z' \leftarrow \arg\max_k s(z^{(k)})$ (5); $x' \leftarrow \mathrm{clip}(z' B^\top)$; $f' \leftarrow f(x')$ ▷ 1
10: $\quad$ **if** $f' < f_z$ or $U(0,1) < p_e$ (6) **then** $z \leftarrow x' B$, $f_z \leftarrow f'$; update S^H, S^M and R
11: $\quad$ **if** $f' < f^\star$ **then** $p^\star \leftarrow x'$; copy x' into a random row of S^H and S^M
12: write S^H, S^M back into X^H, X^M; update $p^H, p^M, p^\star$
13: **if** FAD after Phase 2 **then** FAD stage ▷ N

5. Properties

Proposition 1 (Rotation invariance of the exchange)**.** *Let Q be an orthogonal matrix and consider the rotated problem $\tilde{f}(y) = f(yQ^\top)$. Suppose the populations, subsets and global best of the rotated run are the rotated images $\tilde{X} = XQ$ of those of the original run, the pooled covariance has distinct eigenvalues, the same random numbers are drawn, and no clipping occurs. Then the eigen-frame exchange produces, in every episode, the candidate $\tilde{x}' = x'Q$, with the same objective value and the same acceptance decision.*

Proof. The pooled covariance of the rotated populations is $\tilde{C} = Q^\top C Q$, which has the same eigenvalues as C and, since they are distinct, eigenvectors $\tilde{B} = Q^\top B S$ with S a diagonal matrix of signs, in the same order. The eigen coordinates of a rotated point are therefore $\tilde{z} = xQ\,Q^\top B S = zS$: each coordinate equals the original one up to a fixed sign. Ranks depend only on objective values, which are equal because $\tilde{f}(xQ) = f(x)$. Assembling a candidate by copying coordinate d from a stored solution commutes with multiplication by S, so $\tilde{z}' = z'S$. The terms of the proxy score (5) are a mean rank, a mean of absolute coordinate differences and a Euclidean norm, all unchanged by sign flips, so the same candidate is selected. Mapping back gives $\tilde{x}' = \tilde{z}'\tilde{B}^\top = z'SS^\top B^\top Q = x'Q$, whose value is $\tilde{f}(x'Q) = f(x')$, and the acceptance rule depends only on objective values. □

Remark 1. The axis-frame exchange ($B = I$) has no such property: rotating the problem changes which coordinates are copied together. Proposition 1 concerns the exchange step only. The HBA and MPA moves of Phase 1 are coordinate-wise and not rotation invariant, and neither is clipping to the box, so EIG-MPHBS as a whole is not rotation invariant. A unit test in the repository checks the invariance of the eigen coordinates numerically.

Proposition 2 (No extra evaluations)**.** *In every iteration EIG-MPHBS performs exactly the same number of function evaluations as MPHBS-R, $3N + 1 + E$. The additional arithmetic cost is $O(ND^2 + D^3)$ for the covariance and its eigendecomposition and $O(D^2)$ per episode for the projections, i.e. $O(ND^2 + D^3 + ED^2)$ per iteration.*

Proof. The frame is computed from points already evaluated, and the mediator evaluates one start point and one candidate per episode in either frame. The covariance of $2N$ points costs $O(ND^2)$, the symmetric eigendecomposition $O(D^3)$, and each projection or back-projection of a D-vector $O(D^2)$. □

Overall complexity. Phase 1 and the FAD stage cost $O(ND)$ arithmetic per iteration plus $3N$ evaluations, and the mediator $O(EKD)$ plus $1+E$ evaluations. With T iterations, EIG-MPHBS costs $O\Big(T(ND^2+D^3+EKD+ED^2)\Big)$ arithmetic and $T(3N+1+E)$ evaluations, against $O\Big(T(ND+EKD)\Big)$ and the same number of evaluations for MPHBS-R. For $D \le 31$ and $N = 30$ the extra arithmetic is small compared with the mediator; Section 7.10 measures it. A unit test checks that EIG-MPHBS and MPHBS-R use identical evaluation counts.

6. Experimental setup

6.1. Benchmarks

We use four benchmark sets with 56 instances in total:

- CEC2022 (Kumar et al., 2021), F1–F12, at $D = 10$ and $D = 20$ (24 instances), computed with the official code. The suite contains basic, hybrid and composition functions, all shifted and most rotated.
- BBOB (Hansen et al., 2009), f1–f24 at $D = 20$, with problem instance $((r-1) \bmod 15) + 1$ in run r. The suite covers separable, moderately conditioned, ill-conditioned, multimodal with global structure and multimodal with weak structure functions, and contains separable and rotated versions of the same functions, which allows a direct test of rotation dependence.
- Eight FIR filter-design problems ($D = 31$) introduced with MPHBS (Flores and Olivares, 2026): low-pass, high-pass, band-pass and band-stop filters of order 30, each in an easier and a harder variant. The decision vector is the impulse response $h \in [-1, 1]^{31}$ and the objective is

$$F(h) = W_p \frac{1}{|P|} \sum_{n \in P} \Big(|H(\omega_n)| - 1\Big)^2 + W_s \frac{1}{|S|} \sum_{n \in S} |H(\omega_n)|^2, \tag{8}$$

  where H is the 2048-point discrete Fourier transform of h, P and S are the pass- and stop-band grid points, $W_p = 1$ and $W_s \in \{100, 120\}$. The coefficients interact through the frequency response, so the problems are not separable.

The budget is 2×10^5 FEs on CEC2022 and BBOB and 1.5×10^5 on FIR. Each algorithm is run 30 times per instance. The outcome is the final error $f(x_{\text{best}}) - f^\star$ (the objective value F on FIR, whose optimum is unknown), floored at 10^{-8}.

6.2. Algorithms

Table 2 lists the eleven algorithms and their settings. MPHBS, MPHBS-R and EIG-MPHBS share all code except the flags shown. HBA and MPA, the parent algorithms, were ported from the reference MATLAB code and validated against published per-run results (no significant differences in 80 distribution comparisons on CEC2022 and FIR). PSO, DE, GWO and WOA are four widely used metaheuristics with their standard settings from the original publications, and L-SHADE (Tanabe and Fukunaga, 2014) and CMA-ES with IPOP restarts (Hansen and Ostermeier, 2001; Auger and Hansen, 2005; Hansen et al., 2019) are state-of-the-art references; both were checked against published CEC results before the experiments. All algorithms use the same budget. The classical algorithms stop exactly at the budget, while the MPHBS family stops after the last complete iteration (199 930 of 200 000 evaluations on CEC2022 and BBOB, 149 857 of 150 000 on FIR). No algorithm was tuned on our benchmarks beyond the choice of q described above.

Table 2: Algorithms and settings. N: population size; the HBA–MPA hybrids run two populations of N each.

Algorithm	Settings
EIG-MPHBS	proposed; random mediator and eigen-frame exchange with probability $q = 0.5$; $N = 30$; K, N_i, ρ as in Section 3.4
MPHBS-R	MPHBS with a uniform random mediator at the same budget (control); same settings
MPHBS	SARSA mediator (Flores and Olivares, 2026), learning rate 0.1, discount $\gamma = 0.3$; same settings
HBA	Hashim et al. (2022); $N = 30$, $\beta = 6$, $C = 2$
MPA	Faramarzi et al. (2020); $N = 30$, $P = 0.5$, FADs $= 0.2$
PSO	global best, inertia $0.9 \to 0.4$ (Kennedy and Eberhart, 1995; Shi and Eberhart, 1998); $c_1 = c_2 = 2$, $\lvert v\rvert \le 0.2(u - \ell)$; $N = 30$
DE	DE/rand/1/bin (Storn and Price, 1997); $F = 0.5$, $CR = 0.9$; $N = 30$
GWO	Mirjalili et al. (2014); $a : 2 \to 0$; $N = 30$
WOA	Mirjalili and Lewis (2016); $a : 2 \to 0$, $b = 1$; $N = 30$
L-SHADE	Tanabe and Fukunaga (2014); $N_{\text{init}} = 18D$, $N_{\min} = 4$, $H = 6$, archive rate 2.6, $p = 0.11$
CMA-ES	IPOP-CMA-ES (Auger and Hansen, 2005), pycma (Hansen et al., 2019); $\sigma_0 = 0.3(u - \ell)$, default λ, population doubled at each restart

6.3. Experimental design and statistics

Runs with the same benchmark, instance and run number share their seed across all algorithms, so comparisons are paired. Before any confirmatory run we fixed in a design document the algorithms, the benchmarks and two hypotheses on a held-out set of 28 instances that were not used in any pilot: CEC2022 $D = 10$ F1–F12 and BBOB f2, f4–f7, f9, f11, f13, f14, f17–f20, f22–f24. For each instance we compute the Vargha–Delaney effect size (Vargha and Delaney, 2000) $A = P(e_1 < e_2) + \frac{1}{2}P(e_1 = e_2)$ over all 30×30 pairs of final errors, where e_1 belongs to EIG-MPHBS; $A = 0.5$ means no difference and $A > 0.5$ favours EIG-MPHBS.

- **H1** (RQ1): EIG-MPHBS outperforms MPHBS-R, i.e. the median of A over the held-out instances exceeds 0.5. Since the two differ only in the frame, this isolates the effect of the eigen frame.
- **H2** (RQ1): EIG-MPHBS outperforms MPHBS.

Each is tested with a one-sided Wilcoxon signed-rank test (Wilcoxon, 1945) of $A - 0.5$ over the 28 instances, with Holm correction (Holm, 1979) over the two hypotheses at $\alpha = 0.05$. All other analyses are secondary: the same statistics on all 56 instances; per-instance two-sided paired Wilcoxon tests with Holm correction within each benchmark and comparator; Friedman ranks (Derrac et al., 2011; Demšar, 2006); the rotation pairs of BBOB; per-iteration traces of the exchange; the comparison with EA4eig; and run time. The six classical algorithms were added after the main analysis to provide the comparisons that are customary in this field; their addition is recorded in the design document and does not affect H1 and H2.

6.4. Implementation

All algorithms are implemented in Python with NumPy in one framework that counts every evaluation and records the best-so-far value every 1000 evaluations (750 on FIR) and per-iteration traces of the hybrids. Runs were executed in parallel, one run per core, on a 12-core laptop.

7. Results

7.1. Confirmatory tests (RQ1)

Table 3 and Fig. 2 show the confirmatory results. On the held-out instances, EIG-MPHBS is better than MPHBS-R on 14 instances, tied on 6 and worse on 8 (one-sided $p = 0.031$), and better than MPHBS on 15, tied on 5 and worse on 8 ($p = 0.022$). After Holm correction both hypotheses are supported ($p_{\text{Holm}} = 0.043$ for each). The evidence is not strong, however. The median effect is small (0.500 and 0.514), because most CEC2022 $D = 10$ functions are solved equally well by all three variants, and when the instances with $A = 0.5$ are kept in the test with Pratt or zero-split handling the Holm-adjusted p-values become 0.056 and 0.057. The decision rule fixed in advance used the default handling, which drops these instances; we report the alternatives so that readers can weigh the result.

The per-instance tests show where the effect comes from. Against MPHBS-R, EIG-MPHBS is significantly better on 5 held-out instances (BBOB f6, f9, f11, f14, f19) and worse on 1 (f4, the separable Rastrigin variant). Against MPHBS it is better on 6 and never significantly worse. The gains are concentrated on BBOB functions with ill-conditioned or rotated structure.

On all 56 instances the picture is clearer: the median A is 0.520 against MPHBS-R ($p = 7.7\times10^{-4}$; 12 instances significantly better, 1 worse) and 0.529 against MPHBS ($p = 5.2\times10^{-4}$; 14 better, 0 worse). The largest effects are on FIR (median A 0.703 against MPHBS) and CEC2022 $D = 20$ (0.611). These instances include the pilot instances on which q was chosen, so their effect sizes may be optimistic.

Table 3: Confirmatory tests on the 28 held-out instances (H1, H2) and their repetition on all 56 instances (H1′, H2′, secondary: the 28 additional instances include those used to choose the eigen-frame probability). $\tilde{A}$: median Vargha–Delaney effect over instances (> 0.5 favours EIG-MPHBS). W/T/L: instances with $A > 0.5$, $A = 0.5$, $A < 0.5$. p: one-sided Wilcoxon signed-rank test of $A - 0.5$ over instances; Holm over {H1, H2}. The two sensitivity columns keep the instances with $A = 0.5$ (Pratt and zero-split handling). Sig.: instances where a paired per-instance test (Holm within benchmark) favours EIG-MPHBS / the comparator.

	Comparison	$\tilde{A}$	W/T/L	p	p_{Holm}	p_{Holm} (Pratt)	p_{Holm} (zsplit)	Sig.
H1	EIG-MPHBS vs MPHBS-R	0.500	14/6/8	0.031	0.043	0.056	0.057	5/1
H2	EIG-MPHBS vs MPHBS	0.514	15/5/8	0.022	0.043	0.056	0.057	6/0
H1′	all 56 instances (vs MPHBS-R)	0.520	31/7/18	7.7×10^{-4}	–	–	–	12/1
H2′	all 56 instances (vs MPHBS)	0.529	33/6/17	5.2×10^{-4}	–	–	–	14/0

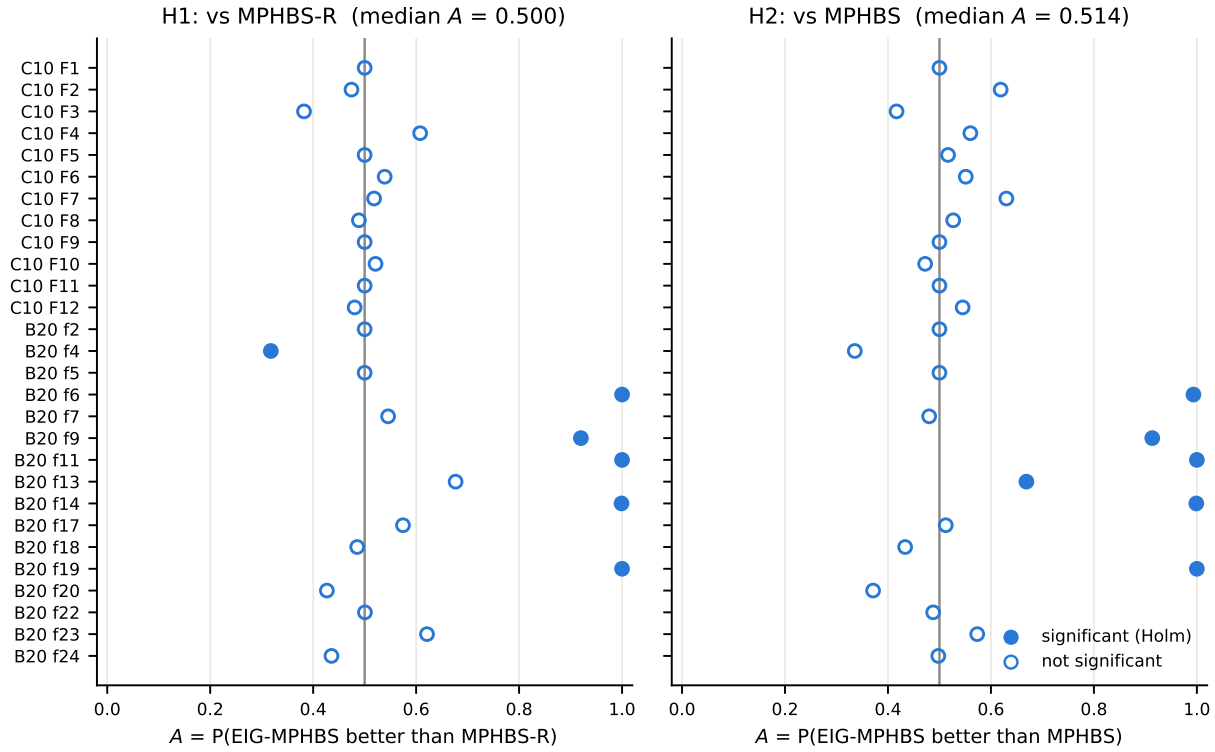


Figure 2: Effect size A of EIG-MPHBS on each of the 28 held-out instances against MPHBS-R (H1) and MPHBS (H2); $A > 0.5$ favours EIG-MPHBS. Filled markers: significant per-instance paired test (Holm within benchmark). C10: CEC2022 $D = 10$; B20: BBOB $D = 20$.

7.2. Rotation dependence (RQ2)

Table 4 compares BBOB functions that differ only by a rotation. On the separable ellipsoid (f2) all algorithms reach the accuracy floor. On the rotated ellipsoid (f10), MPHBS and MPHBS-R stall at median errors of 121 and 89.9, while EIG-MPHBS reaches 8.47, a 10.6-fold reduction relative to MPHBS-R and 14.3-fold relative to MPHBS. This is the case the method targets. The gap to CMA-ES and L-SHADE, which solve f10, remains, because Phase 1 of the hybrid is still coordinate-wise. On Rastrigin, the rotated version (f15) is hard for all three hybrids and the eigen frame does not help (median 54.2 against 58.7 for MPHBS-R); on the separable version (f3) EIG-MPHBS is slightly worse (1.99 against 0.995), as expected when half of the exchanges no longer follow the axes along which the function separates. Highly multimodal rotated problems need more than a change of frame in the exchange step.

Table 4: Median final error on BBOB function pairs that differ only by a rotation ($D = 20$, 30 runs; values at or below 10^{-8} shown as 0).

	Ellipsoid		Rastrigin	
Algorithm	f2 (separable)	f10 (rotated)	f3 (separable)	f15 (rotated)
EIG-MPHBS	0	8.47	1.99	54.2
MPHBS-R	0	89.9	0.995	58.7
MPHBS	0	121	0.995	51.2
CMA-ES	0	0	8.95	2.98
L-SHADE	0	0	0	4.13

7.3. Why the eigen frame helps: exchange acceptance and diversity (RQ2)

Fig. 3 shows, for one separable and three coupled BBOB functions, the fraction of exchanged candidates that are accepted in each iteration and the diversity of the populations (mean distance to the centroid, normalised by the box width), as medians over 30 runs. On the separable ellipsoid f2, the three hybrids behave alike: many candidates are accepted at first and acceptance then drops as the populations converge. On the rotated ellipsoid f10 the axis-frame exchange almost stops working after the first few thousand evaluations: in the first quarter of the budget, 0.4% of the candidates of MPHBS-R and 0.6% of those of MPHBS are accepted, against 4.4% for EIG-MPHBS. The discus f11 shows the same pattern (0.0% and 0.0% against 1.0%). Because the exchanged candidates now stay inside the valley, the populations of EIG-MPHBS contract faster along it, which is visible as a steeper decline in diversity on f10 and f11. On f14 (sum of different powers, moderately coupled) the acceptance rates are similar and the effect is smaller. This supports the explanation of Section 4: the eigen frame restores the exchange channel on coupled problems rather than changing the native moves.

7.4. Comparison with ten algorithms (RQ3)

Fig. 4 and Table 5 give the mean Friedman ranks of the eleven algorithms, and Table 6 and Fig. 5 the per-instance significance of EIG-MPHBS against each of them. Tables 10–17

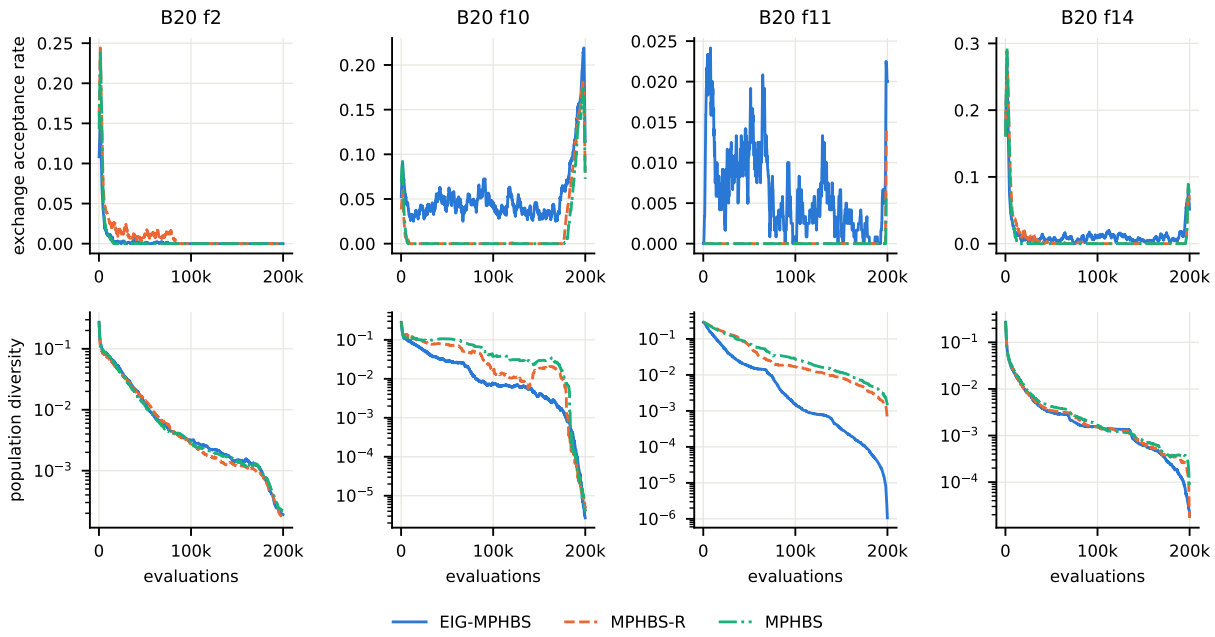


Figure 3: Mechanism of the eigen frame on BBOB $D = 20$. Top: fraction of exchanged candidates accepted per iteration (moving average over 25 iterations). Bottom: population diversity (log scale). Medians over 30 runs. f2 is separable; f10 (ellipsoid), f11 (discus) and f14 (sum of different powers) are rotated.

in Section 7.11 give medians, means and standard deviations per instance. The Friedman test rejects equal performance on every benchmark ($p \leq 1.4{\times}10^{-12}$).

EIG-MPHBS ranks 3 of 11 overall (mean rank 4.01), behind L-SHADE (2.60) and CMA-ES (2.97) and ahead of MPHBS-R (4.62), MPHBS (4.64) and all classical metaheuristics. It has the best mean rank on CEC2022 $D = 10$ and ranks third on the other three benchmarks.

- **Against its own family**, EIG-MPHBS is significantly better than MPHBS on 14 instances and worse on none, and better than MPHBS-R on 12 and worse on 1.
- **Against the parent algorithms**, it is significantly better than HBA on 47 of 56 instances and worse on none, and better than MPA on 24 and worse on 6; MPA is the strongest of the classical algorithms, and many of the ties with it are CEC2022 $D = 10$ functions that both solve.
- **Against PSO, DE, GWO and WOA**, it is better on 44, 37, 52 and 56 instances and worse on 1, 6, 0 and 0. Over the six classical algorithms it wins 260 and loses 13 of 336 comparisons.
- **Against L-SHADE and CMA-ES**, it is better on 4 and 8 instances and worse on 34 and 31. On CEC2022 $D = 10$ the three are close, but at $D = 20$, on BBOB and on FIR the two references are clearly stronger.

Fig. 5 shows that the losses to L-SHADE and CMA-ES are spread over all benchmarks except CEC2022 $D = 10$, while the wins over the classical algorithms are nearly uniform. GWO and WOA perform poorly on these shifted problems; both are known to be biased towards the centre of the search space (Kůdela, 2022), which our unit tests confirm on a shifted sphere.

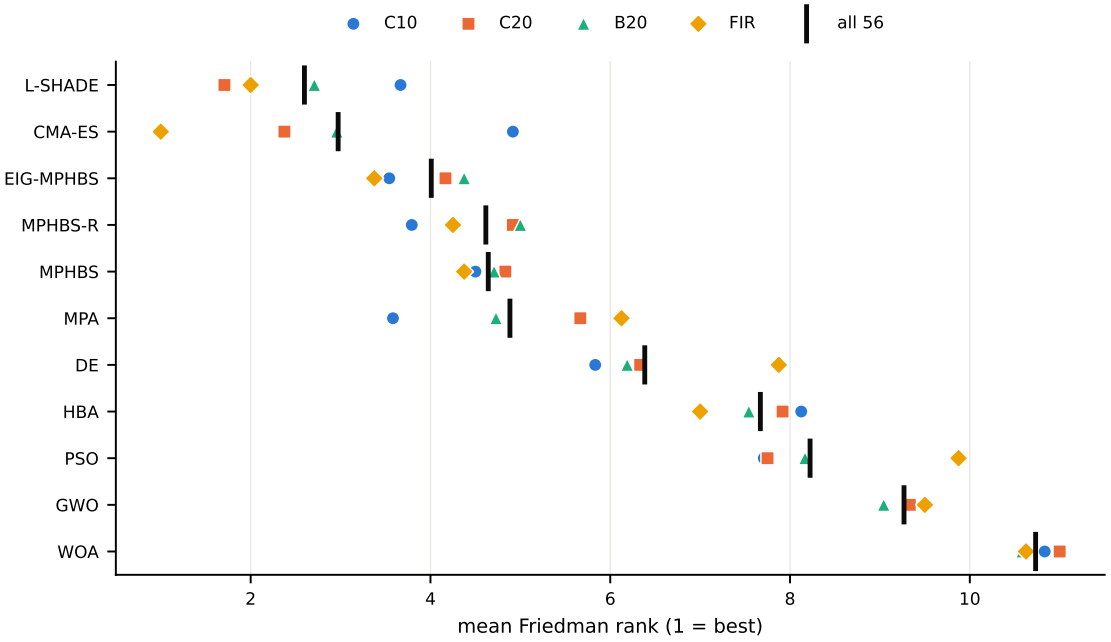


Figure 4: Mean Friedman rank of each algorithm per benchmark (markers) and over all 56 instances (vertical bar); 1 = best. C10/C20: CEC2022 $D = 10/20$; B20: BBOB $D = 20$.

Table 5: Mean Friedman rank of the eleven algorithms (rank of the median final error on each instance, 1 = best; lowest in bold), per benchmark and over all 56 instances. Last row: Friedman test p per benchmark.

Algorithm	C10 (12)	C20 (12)	B20 (24)	FIR (8)	All (56)
L-SHADE	3.67	**1.71**	**2.71**	2.00	**2.60**
CMA-ES	4.92	2.38	2.96	**1.00**	2.97
EIG-MPHBS	**3.54**	4.17	4.38	3.38	4.01
MPHBS-R	3.79	4.92	5.00	4.25	4.62
MPHBS	4.50	4.83	4.71	4.38	4.64
MPA	3.58	5.67	4.73	6.12	4.88
DE	5.83	6.33	6.19	7.88	6.38
HBA	8.12	7.92	7.54	7.00	7.67
PSO	7.71	7.75	8.17	9.88	8.22
GWO	9.50	9.33	9.04	9.50	9.27
WOA	10.83	11.00	10.58	10.62	10.73
Friedman p	3.9×10^{-14}	1.3×10^{-14}	2.8×10^{-26}	1.4×10^{-12}	–

Table 6: EIG-MPHBS against each algorithm: number of instances where it is significantly better / not significantly different / significantly worse (paired two-sided Wilcoxon signed-rank test over the 30 seed-paired runs, Holm within each benchmark and comparator, $\alpha = 0.05$).

vs	C10 (12)	C20 (12)	B20 (24)	FIR (8)	Total (56)
L-SHADE	1/9/2	0/4/8	3/5/16	0/0/8	4/18/34
CMA-ES	3/7/2	1/5/6	4/5/15	0/0/8	8/17/31
MPHBS-R	0/12/0	3/9/0	7/16/1	2/6/0	12/43/1
MPHBS	0/12/0	2/10/0	7/17/0	5/3/0	14/42/0
MPA	1/11/0	5/5/2	10/10/4	8/0/0	24/26/6
DE	9/2/1	7/3/2	13/8/3	8/0/0	37/13/6
HBA	10/2/0	10/2/0	19/5/0	8/0/0	47/9/0
PSO	9/3/0	9/3/0	18/5/1	8/0/0	44/11/1
GWO	12/0/0	12/0/0	20/4/0	8/0/0	52/4/0
WOA	12/0/0	12/0/0	24/0/0	8/0/0	56/0/0

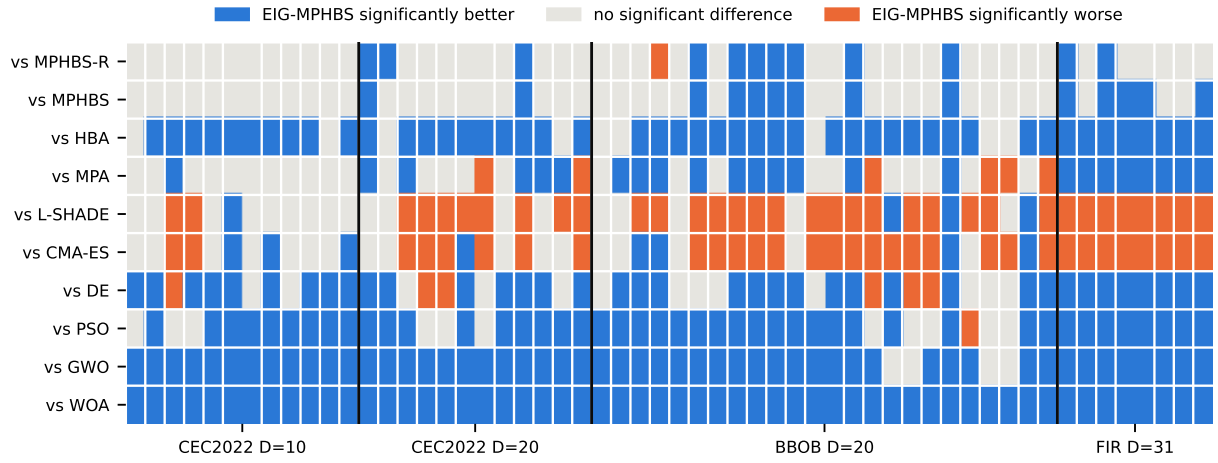


Figure 5: Per-instance comparison of EIG-MPHBS with each algorithm (paired Wilcoxon over 30 runs, Holm within benchmark and comparator). Each column is one instance: CEC2022 F1–F12 at $D = 10$ and $D = 20$, BBOB f1–f24, FIR cases 1–8.

7.5. Distributions of the final error

Fig. 6 shows the distributions of the final error of all eleven algorithms on the two instances per benchmark where EIG-MPHBS differs most from MPHBS (C10 F7, C10 F2, C20 F1, C20 F9, B20 f19, B20 f11, FIR Case 3, FIR Case 1). The spread of EIG-MPHBS is generally narrower than that of MPHBS and MPHBS-R on these instances, and on BBOB f11 and FIR case 3 its whole distribution lies below theirs. The boxplots also show how far the classical metaheuristics fall behind on the shifted and rotated problems, and the large spread of DE and PSO.

7.6. Convergence behaviour

Fig. 7 shows median convergence curves of seven algorithms on six instances, and Figs. 8 and 9 (and Fig. 11 in Section 7.11) on all instances of CEC2022 $D = 20$ and FIR. On

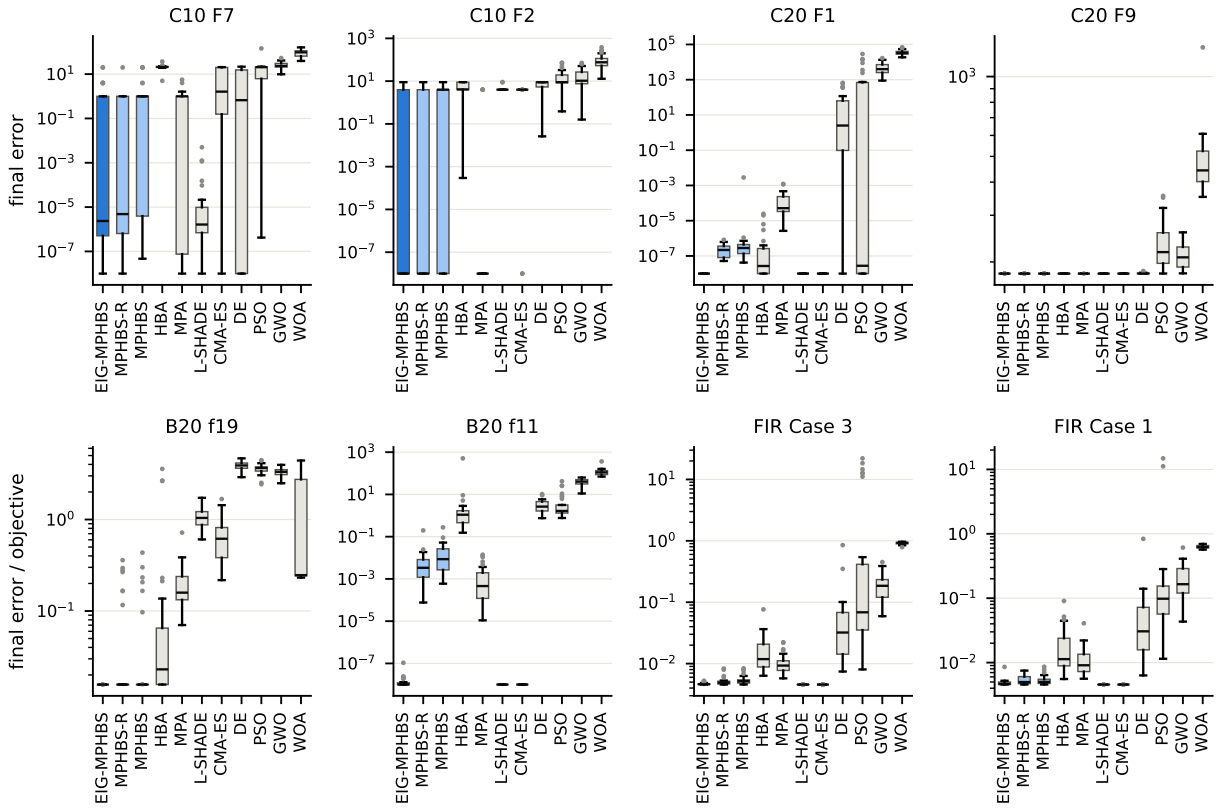


Figure 6: Final error over 30 runs (log scale) on two instances per benchmark. EIG-MPHBS in dark blue, MPHBS-R and MPHBS in light blue, the other algorithms in grey.

BBOB f10, f14 and f19 and on CEC2022 $D = 20$ F1, EIG-MPHBS separates from MPHBS and MPHBS-R early and keeps its lead. On f4 the eigen frame slows it down slightly. On CEC2022 $D = 20$ the hybrid family converges faster than MPA and DE on most functions, while CMA-ES converges fastest on the unimodal ones. On FIR, EIG-MPHBS converges to lower values than MPHBS and MPHBS-R on most cases, but CMA-ES and L-SHADE reach lower values.

7.7. Engineering application: FIR filter design

The FIR problems are the application part of the benchmark. On them EIG-MPHBS is significantly better than MPHBS on five of eight cases and worse on none (Table 6), and the median A against MPHBS is 0.703, the largest of the four benchmarks. Fig. 10 shows the magnitude responses of the median-quality filters of EIG-MPHBS, MPHBS and L-SHADE on the four harder cases. All three meet the pass-band specification closely; the differences lie in the stop-band attenuation, where EIG-MPHBS typically attains a few decibels more attenuation than MPHBS near the transition band, and L-SHADE more than both away from it. In practical terms, all three produce usable filters, and the objective differences of Table 13 correspond to a few decibels of stop-band attenuation.

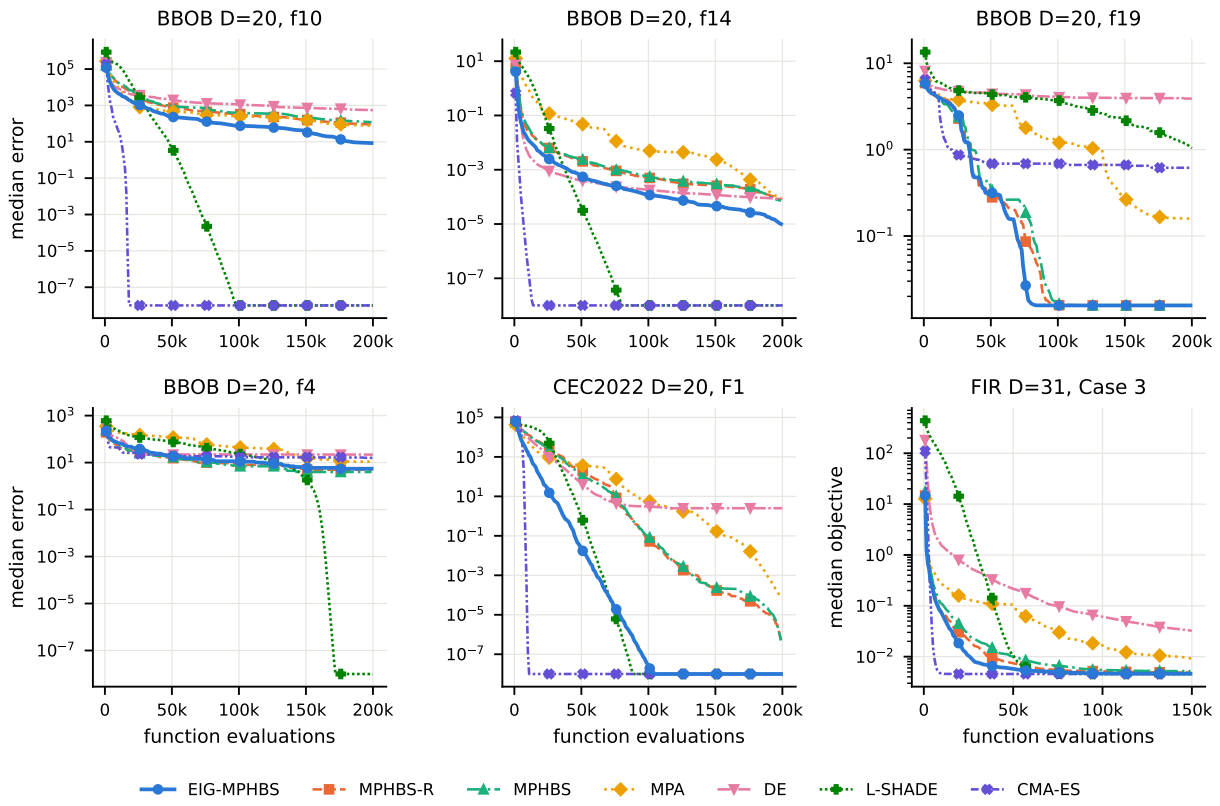


Figure 7: Median best-so-far error over 30 runs against function evaluations (FIR: objective value), log scale. The CEC2022 and FIR instances are those with the strongest per-instance effect of EIG-MPHBS against MPHBS-R; BBOB f4 is the held-out instance where it is significantly worse.

7.8. Ablation and sensitivity to the frame probability (RQ4)

Table 7 reports the pilot study that preceded the main experiments: four variants of the hybrid on 28 instances, 10 seed-paired runs each. Removing the exchange altogether ("no exchange") is clearly harmful on FIR and CEC2022 $D = 20$, which confirms that the exchange is the important component of the hybrid. Among the frame probabilities, $q = 0.5$ has the best mean rank on CEC2022 $D = 20$ and FIR and ties with $q = 1$ on the eight BBOB pilot instances. Always using the eigen frame ($q = 1$) loses some of the benefit on FIR, which has partly separable structure, and $q = 0$ (the axis frame) is the weakest of the exchange variants on CEC2022 $D = 20$. With 10 runs few per-instance differences are significant, so this table supports the choice of $q = 0.5$ but does not show that it is optimal; an adaptive choice is a natural extension.

7.9. Comparison with EA4eig

EA4eig (Bujok and Kolenovský, 2022) combines eigen crossover with four strong algorithms and was among the best entries of the CEC2022 competition. We compare EIG-MPHBS with the official EA4eig results at the competition budgets (30 runs at 2×10^5 evaluations for $D = 10$ and 10^6 for $D = 20$; EIG-MPHBS was rerun at 10^6 for $D = 20$). The comparison is unpaired and descriptive, because hardware, language and implementation differ. Table 8 shows that EA4eig is significantly better on 8 of 12 functions at $D = 10$ and 10

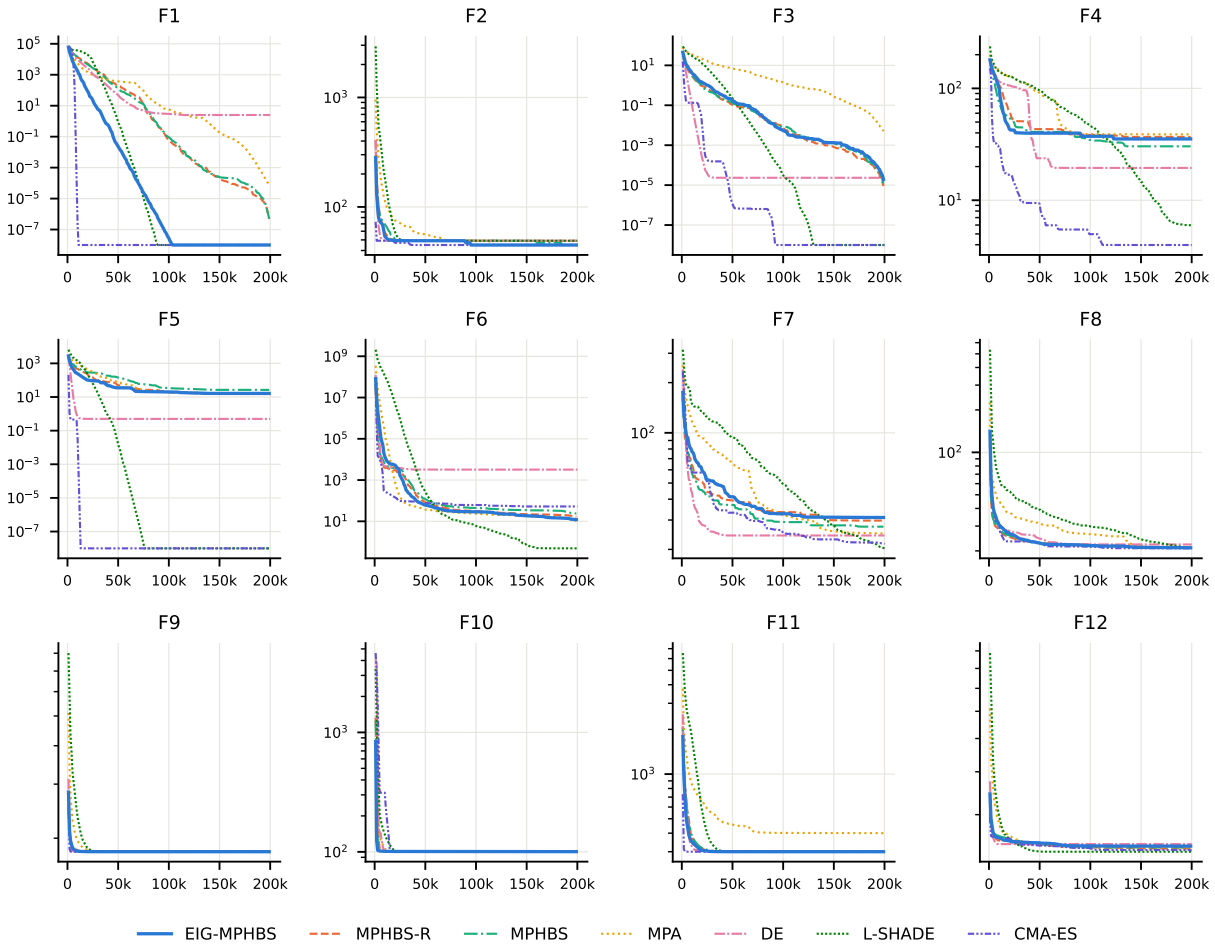


Figure 8: Median best-so-far error on CEC2022 $D = 20$, F1–F12 (30 runs, log scale).

Table 7: Ablation and sensitivity to the eigen-frame probability q in the pilot study (10 seed-paired runs per instance; exploratory, and these instances are excluded from the confirmatory test). "No exchange" removes Phase 2; $q = 0$ is MPHBS-R; $q = 0.5$ is EIG-MPHBS; $q = 1$ always uses the eigen frame. Left: mean rank of the median final error (1 = best). Right: median Vargha–Delaney A of $q = 0.5$ against $q = 0$ and $q = 1$, with the number of instances where it is significantly better / not different / worse (paired Wilcoxon, Holm within benchmark). *BBOB f1, f3, f8, f10, f12, f15, f16, f21.

	Mean rank				$q = 0.5$ against	
Benchmark	no exchange	$q = 0$	$q = 0.5$	$q = 1$	$q = 0$	$q = 1$
B20* (8)	2.19	2.94	2.44	2.44	0.51 (2/6/0)	0.47 (1/7/0)
C20 (12)	2.67	2.92	1.88	2.54	0.64 (1/11/0)	0.61 (1/11/0)
FIR (8)	4.00	2.25	1.38	2.38	0.65 (1/7/0)	0.69 (1/7/0)

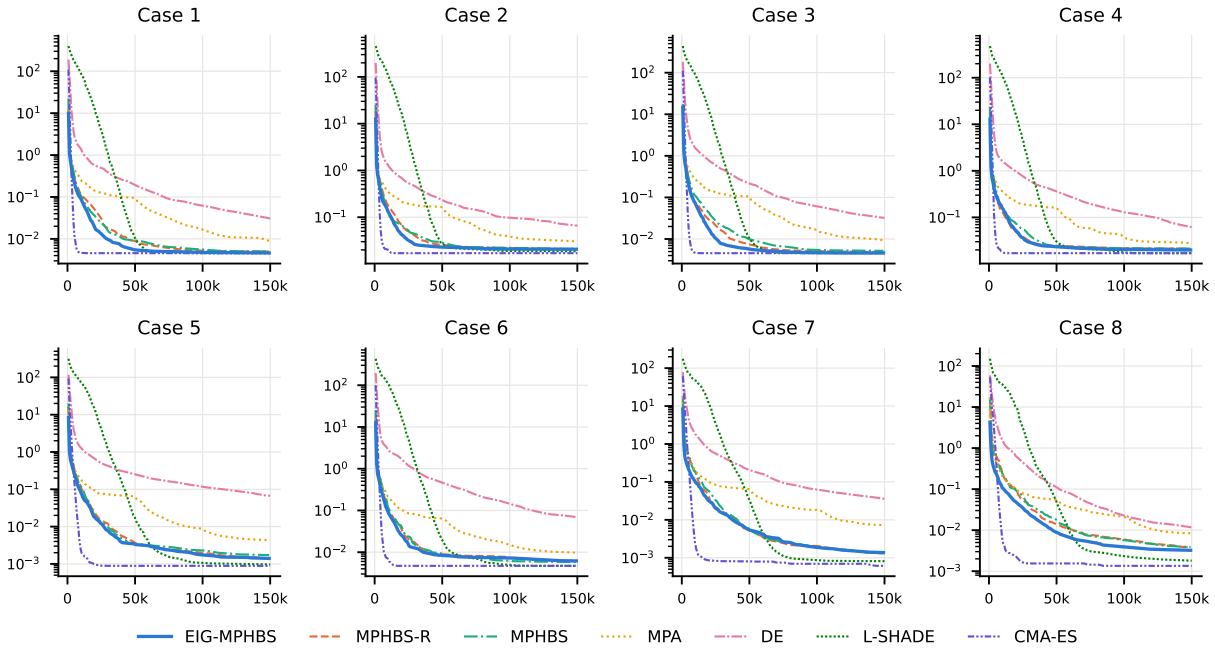


Figure 9: Median best-so-far objective value on the eight FIR filter-design problems (30 runs, log scale).

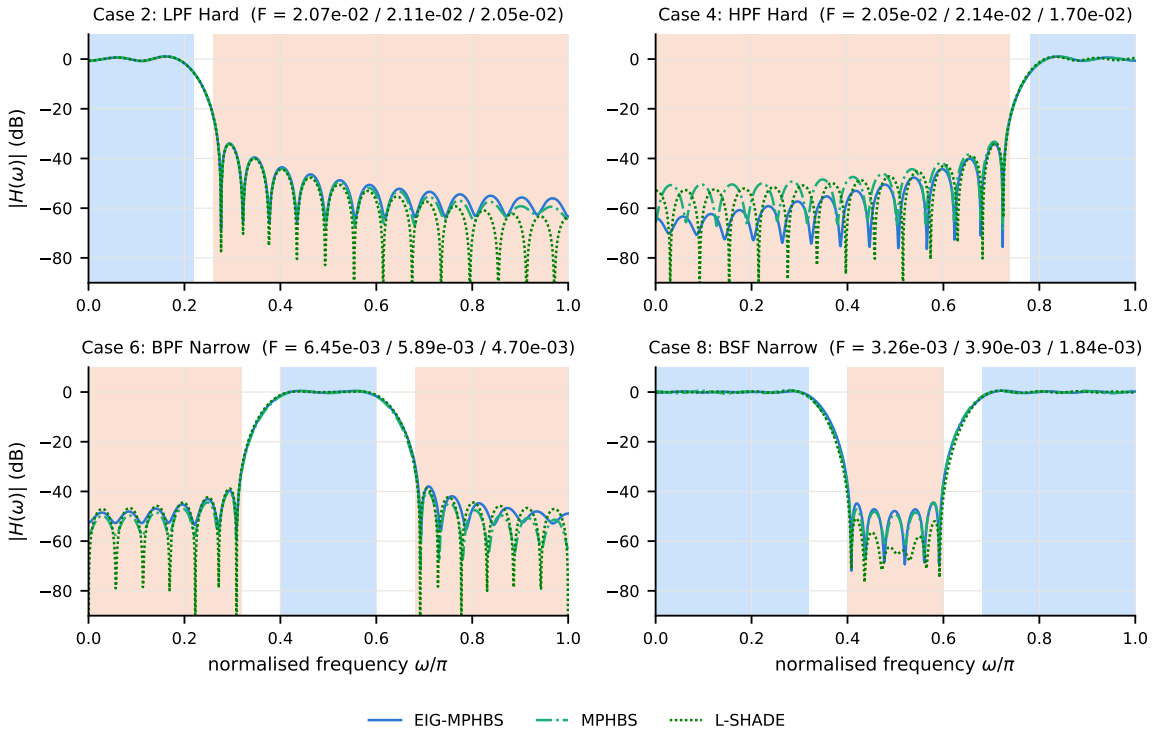


Figure 10: Magnitude response of the median-quality filter (15th of 30 runs by objective value) of three algorithms on the four harder FIR cases. Shaded: pass band (blue) and stop band (orange). Objective values F in the panel titles in the order EIG-MPHBS / MPHBS / L-SHADE.

of 12 at $D = 20$, with no significant difference on the rest; EIG-MPHBS is never significantly better. Adding an eigen-frame exchange to an HBA–MPA hybrid does not close the gap to a cooperative model of state-of-the-art evolutionary algorithms.

Table 8: EIG-MPHBS against the official CEC2022 runs of EA4eig at the competition budgets (2×10^5 evaluations for $D = 10$, 10^6 for $D = 20$; 30 runs each). Median final error (at or below 10^{-8} shown as 0) and Vargha–Delaney A (> 0.5 favours EIG-MPHBS). Superscripts: two-sided Mann–Whitney test, Holm over the 12 functions of each dimension ($^+$ EIG-MPHBS better, $^-$ worse, $^\approx$ no significant difference). Unpaired and descriptive: hardware and implementation differ.

	$D = 10$			$D = 20$		
	EIG-MPHBS	EA4eig	A	EIG-MPHBS	EA4eig	A
F1	$0^\approx$	0	0.50	$0^\approx$	0	0.50
F2	$0^\approx$	0	0.46	47^-	0	0.17
F3	2.27e-7$^-$	0	0.13	7.77e-8$^-$	0	0.15
F4	7.96^-	0.995	0.01	34.8^-	6.96	0.00
F5	$0^\approx$	0	0.50	14.1^-	0	0.00
F6	0.155^-	5.41e-3	0.04	3.28^-	0.106	0.01
F7	2.36e-6$^-$	0	0.02	25.3^-	2.3	0.03
F8	0.16^-	0.0467	0.27	20.6^-	20.3	0.29
F9	229^-	186	0.00	181^-	165	0.00
F10	100^-	100	0.11	$100^\approx$	100	0.34
F11	$0^\approx$	0	0.50	300^-	300	0.16
F12	159^-	146	0.00	240^-	200	0.00

7.10. Computational cost (RQ4)

EIG-MPHBS and MPHBS-R used identical evaluation counts on every one of their paired runs, as Proposition 2 states. Table 9 reports the run times. The eigen frame adds 3.2–5.5% to the median run time of MPHBS-R, and EIG-MPHBS is faster than MPHBS, whose SARSA bookkeeping costs more than the eigendecomposition. The classical algorithms are faster per run because their iterations are simpler; with expensive objectives, where evaluations dominate, the differences in algorithmic overhead become negligible.

7.11. Detailed per-instance results

For completeness, Fig. 11 shows the convergence curves on CEC2022 $D = 10$, and Tables 10–13 and 14–17 give the median and the mean (standard deviation) of the final error of all eleven algorithms on every instance. The superscripts in the median tables mark the per-instance tests summarised in Table 6 and Fig. 5.

Table 9: Median wall-clock time per run in seconds (Python 3, NumPy, one core per run, 11 runs in parallel on a 12-core laptop; other jobs shared the machine, so times are indicative). Last row: median paired ratio of the run times of EIG-MPHBS and MPHBS-R, which use identical evaluation counts.

Algorithm	C10	C20	B20	FIR
EIG-MPHBS	61.3	60.9	133.2	79.8
MPHBS-R	60.6	59.2	132.1	77.1
MPHBS	81.1	75.9	158.3	86.9
HBA	7.1	8.4	8.4	20.5
MPA	4.5	5.7	5.7	19.2
L-SHADE	19.4	11.6	23.4	24.9
CMA-ES	120.8	89.1	183.2	75.6
DE	17.6	18.2	18.3	28.4
PSO	4.9	6.1	5.9	19.3
GWO	5.7	7.2	7.0	18.9
WOA	5.5	6.6	6.4	17.7
EIG-MPHBS / MPHBS-R	1.055	1.033	1.032	1.032

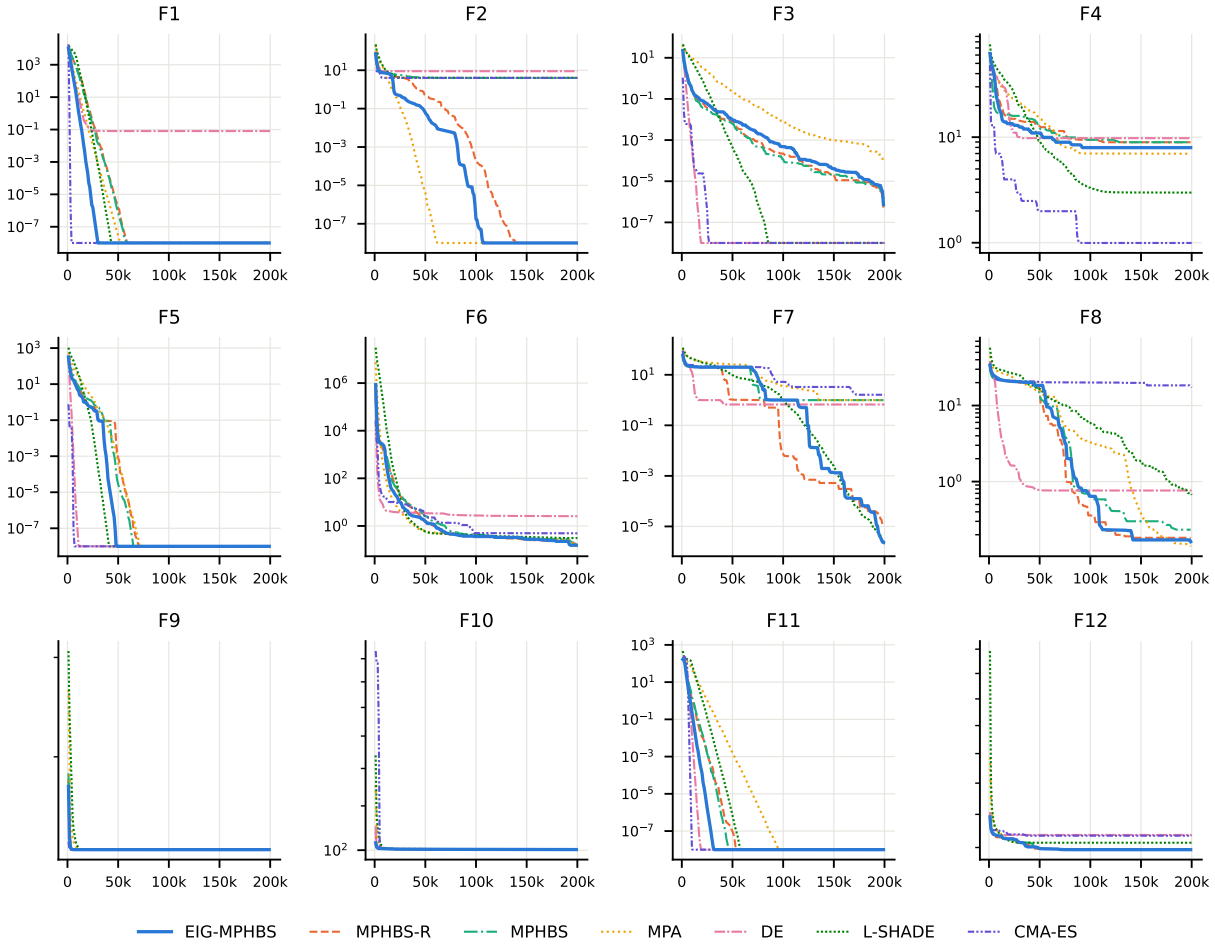


Figure 11: Median best-so-far error on CEC2022 $D = 10$, F1–F12 (30 runs, log scale).

Table 10: Median final error over 30 runs on CEC2022 D=10 (FIR: objective value). Errors at or below 10^{-8} are shown as 0; lowest median in bold. Superscripts compare with EIG-MPHBS (paired Wilcoxon, Holm within benchmark and comparator): $^{+}$ EIG-MPHBS significantly better, $^{-}$ significantly worse, $^{\approx}$ no significant difference.

	EIG-MPHBS	MPHBS-R	MPHBS	HBA	MPA	L-SHADE	CMA-ES	DE	PSO	GWO	WOA
F1	**0**	**0**$^{\approx}$	**0**$^{\approx}$	**0**$^{\approx}$	**0**$^{\approx}$	**0**$^{\approx}$	**0**$^{\approx}$	0.0805^{+}	**0**$^{\approx}$	137^{+}	8.29e+03^{+}
F2	**0**	**0**$^{\approx}$	3.99$^{\approx}$	4.09^{+}	**0**$^{\approx}$	3.99$^{\approx}$	3.99$^{\approx}$	8.92^{+}	8.92^{+}	10.4^{+}	76^{+}
F3	2.27e-7	2.46e-8$^{\approx}$	9.39e-8$^{\approx}$	5.75e-5^{+}	2.98e-5^{+}	**0**$^{-}$	**0**$^{-}$	**0**$^{-}$	**0**$^{\approx}$	0.131^{+}	34.8^{+}
F4	7.96	8.95$^{\approx}$	8.95$^{\approx}$	15.9^{+}	6.96$^{\approx}$	2.98^{-}	**0.995**$^{-}$	9.79^{+}	10.4$^{\approx}$	14.1^{+}	39^{+}
F5	**0**	**0**$^{\approx}$	**0**$^{\approx}$	0.544^{+}	**0**$^{\approx}$	**0**$^{\approx}$	**0**$^{\approx}$	**0**$^{+}$	**0**$^{+}$	1.36^{+}	402^{+}
F6	0.155	0.178$^{\approx}$	0.169$^{\approx}$	1.42e+03^{+}	**0.138**$^{\approx}$	0.311^{+}	0.49^{+}	2.56^{+}	3.35e+03^{+}	2.9e+03^{+}	2.25e+03^{+}
F7	2.36e-6	4.83e-6$^{\approx}$	0.995$^{\approx}$	21^{+}	0.995$^{\approx}$	**1.63e-6**$^{\approx}$	1.62$^{\approx}$	0.667$^{\approx}$	20.6^{+}	23.3^{+}	96.2^{+}
F8	0.16	0.18$^{\approx}$	0.232$^{\approx}$	21^{+}	**0.138**$^{\approx}$	0.669$^{\approx}$	17.3^{+}	0.763^{+}	20.7^{+}	24.5^{+}	32.4^{+}
F9	**229**	**229**$^{\approx}$	**229**$^{\approx}$	230^{+}	**229**$^{\approx}$	**229**$^{\approx}$	**229**$^{\approx}$	**229**$^{\approx}$	230^{+}	231^{+}	388^{+}
F10	100	100$^{\approx}$	100$^{\approx}$	210^{+}	100$^{\approx}$	**100**$^{\approx}$	100$^{\approx}$	100^{+}	210^{+}	207^{+}	253^{+}
F11	**0**	**0**$^{\approx}$	**0**$^{\approx}$	**0**$^{\approx}$	**0**$^{\approx}$	**0**$^{\approx}$	**0**$^{\approx}$	**0**$^{+}$	178^{+}	130^{+}	279^{+}
F12	159	**159**$^{\approx}$	160$^{\approx}$	183^{+}	160$^{\approx}$	161$^{\approx}$	163^{+}	164^{+}	171^{+}	164^{+}	233^{+}

Table 11: Median final error over 30 runs on CEC2022 D=20 (FIR: objective value). Errors at or below 10^{-8} are shown as 0; lowest median in bold. Superscripts compare with EIG-MPHBS (paired Wilcoxon, Holm within benchmark and comparator): $^{+}$ EIG-MPHBS significantly better, $^{-}$ significantly worse, $^{\approx}$ no significant difference.

	EIG-MPHBS	MPHBS-R	MPHBS	HBA	MPA	L-SHADE	CMA-ES	DE	PSO	GWO	WOA
F1	**0**	2.18e-7^{+}	2.82e-7^{+}	2.69e-8^{+}	5.13e-5^{+}	**0**$^{\approx}$	**0**$^{\approx}$	2.51^{+}	2.77e-8^{+}	3.97e+03^{+}	3.30e4^{+}
F2	44.9	49.1^{+}	44.9$^{\approx}$	49.1$^{\approx}$	44.9$^{\approx}$	49.1$^{\approx}$	**44.9**$^{\approx}$	49.1^{+}	54.4^{+}	93^{+}	653^{+}
F3	1.72e-5	5.03e-6$^{\approx}$	6.93e-6$^{\approx}$	1.09^{+}	4.68e-3^{+}	**0**$^{-}$	**0**$^{-}$	2.33e-5$^{\approx}$	0.556^{+}	2.29^{+}	71.7^{+}
F4	35.3	36.8$^{\approx}$	30.3$^{\approx}$	50.7^{+}	38.8$^{\approx}$	5.97^{-}	**3.98**$^{-}$	19.5^{-}	38$^{\approx}$	44.3^{+}	148^{+}
F5	16.3	15.4$^{\approx}$	26.5$^{\approx}$	161^{+}	14.9$^{\approx}$	**0**$^{-}$	**0**$^{-}$	0.499^{-}	8.15$^{\approx}$	158^{+}	2.32e+03^{+}
F6	12.2	16.8$^{\approx}$	24$^{\approx}$	2.46e+03^{+}	10.6$^{\approx}$	**0.489**$^{-}$	52.5^{+}	3.21e+03^{+}	2.45e+03^{+}	2.08e4^{+}	7.11e7^{+}
F7	31.1	29.8$^{\approx}$	27.4$^{\approx}$	69.1^{+}	24.9^{-}	**20.3**$^{-}$	21.6^{-}	24.2$^{\approx}$	29.2$^{\approx}$	51.8^{+}	200^{+}
F8	21	21$^{\approx}$	21.2$^{\approx}$	24.9^{+}	21.1$^{\approx}$	20.9$^{\approx}$	**20.8**$^{\approx}$	22.2^{+}	41.3^{+}	28.3^{+}	58.2^{+}
F9	181	181^{+}	181^{+}	181^{+}	181^{+}	**181**$^{-}$	**181**$^{-}$	181^{+}	218^{+}	208^{+}	443^{+}
F10	100	100$^{\approx}$	100$^{\approx}$	839^{+}	101^{+}	**100**$^{\approx}$	100$^{\approx}$	101^{+}	292^{+}	912^{+}	3.26e+03^{+}
F11	300	300$^{\approx}$	300$^{\approx}$	300$^{\approx}$	400^{+}	**300**$^{-}$	300$^{\approx}$	303$^{\approx}$	966^{+}	748^{+}	2.9e+03^{+}
F12	243	239$^{\approx}$	241$^{\approx}$	357^{+}	241^{-}	**234**$^{-}$	236^{-}	246^{+}	285^{+}	261^{+}	504^{+}

Table 12: Median final error over 30 runs on BBOB D=20 (FIR: objective value). Errors at or below 10^{-8} are shown as 0; lowest median in bold. Superscripts compare with EIG-MPHBS (paired Wilcoxon, Holm within benchmark and comparator): $^{+}$ EIG-MPHBS significantly better, $^{-}$ significantly worse, $^{\approx}$ no significant difference.

	EIG-MPHBS	MPHBS-R	MPHBS	HBA	MPA	L-SHADE	CMA-ES	DE	PSO	GWO	WOA
f1	**0**	**0**$^{\approx}$	**0**$^{\approx}$	**0**$^{\approx}$	**0**$^{\approx}$	**0**$^{\approx}$	**0**$^{\approx}$	**0**$^{\approx}$	1.19^{+}	0.536^{+}	15.3^{+}
f2	**0**	**0**$^{\approx}$	**0**$^{\approx}$	**0**$^{\approx}$	3.77e-4^{+}	**0**$^{\approx}$	**0**$^{\approx}$	0.295^{+}	3.46e+03^{+}	1.50e4^{+}	1.78e5^{+}
f3	1.99	0.995$^{\approx}$	0.995$^{\approx}$	35.3^{+}	5.47^{+}	**0**$^{-}$	8.95^{+}	13.1^{+}	19^{+}	45.6^{+}	337^{+}
f4	5.47	4.97^{-}	3.98$^{\approx}$	45.3^{+}	10.9^{+}	**0**$^{-}$	15.9^{+}	21.9^{+}	17.9^{+}	66.2^{+}	357^{+}
f5	**0**	**0**$^{\approx}$	**0**$^{\approx}$	92.1^{+}	**0**$^{\approx}$	**0**$^{\approx}$	**0**$^{\approx}$	**0**$^{\approx}$	30.8^{+}	10.8^{+}	106^{+}
f6	3.26e-8	1.78e-6^{+}	3.37e-6^{+}	5.41e-4^{+}	1.06e-4^{+}	**0**$^{-}$	**0**$^{-}$	**0**$^{\approx}$	114^{+}	40.8^{+}	1.19e+03^{+}
f7	4.31	3.59$^{\approx}$	2.62$^{\approx}$	7.55^{+}	2.05$^{\approx}$	**0**$^{-}$	**0**$^{-}$	2.62$^{\approx}$	13.1^{+}	14.5^{+}	183^{+}
f8	3.64e-7	4.22e-4^{+}	1.84e-3^{+}	3.31^{+}	10.7^{+}	**0**$^{-}$	**0**$^{-}$	17.2^{+}	12.8^{+}	90.7^{+}	2.82e+03^{+}
f9	7.03e-3	0.798^{+}	1.48^{+}	6.41^{+}	4.57^{+}	**0**$^{-}$	**0**$^{-}$	17.7^{+}	16.9^{+}	22^{+}	107^{+}
f10	8.47	89.9^{+}	121^{+}	829^{+}	77.3^{+}	**0**$^{-}$	**0**$^{-}$	550^{+}	7.09e+03^{+}	1.15e4^{+}	2.28e5^{+}
f11	**0**	3.36e-3^{+}	8.62e-3^{+}	1.09^{+}	4.60e-4^{+}	**0**$^{\approx}$	**0**$^{\approx}$	2.65^{+}	1.65^{+}	40.4^{+}	111^{+}
f12	4.65	3.47$^{\approx}$	4.35$^{\approx}$	3.45$^{\approx}$	5.16$^{\approx}$	**0**$^{-}$	**0**$^{-}$	1.96$^{\approx}$	7.93e4^{+}	2.46e5^{+}	2.85e7^{+}
f13	0.774	2.82$^{\approx}$	2.86$^{\approx}$	3^{+}	1.13$^{\approx}$	**0**$^{-}$	**0**$^{-}$	2.04^{+}	200^{+}	179^{+}	858^{+}
f14	9.73e-6	7.96e-5^{+}	7.21e-5^{+}	7.79e-5^{+}	9.53e-5^{+}	**0**$^{-}$	**0**$^{-}$	8.40e-5^{+}	1.58^{+}	0.26^{+}	8.69^{+}
f15	54.2	58.7$^{\approx}$	51.2$^{\approx}$	70.1^{+}	37.3^{-}	4.13^{-}	**2.98**$^{-}$	19^{-}	57.1$^{\approx}$	63.9^{+}	359^{+}
f16	1.84	1.99$^{\approx}$	1.92$^{\approx}$	6.02^{+}	1.39$^{\approx}$	3.83^{+}	**0.0403**$^{-}$	9.82^{+}	3.21^{+}	1.28$^{\approx}$	15.8^{+}
f17	0.488	0.592$^{\approx}$	0.448$^{\approx}$	1.38^{+}	0.271$^{\approx}$	**0**$^{-}$	**0**$^{-}$	0.0333^{-}	0.753$^{\approx}$	0.918$^{\approx}$	7.25^{+}
f18	1.04	1.15$^{\approx}$	0.86$^{\approx}$	4.83^{+}	0.997$^{\approx}$	4.66e-8^{-}	**0**$^{-}$	0.241^{-}	1.62$^{\approx}$	2.31^{+}	26.9^{+}
f19	**0.0157**	0.0157^{+}	0.0157^{+}	0.023^{+}	0.159^{+}	1.04^{+}	0.616^{+}	3.9^{+}	3.65^{+}	3.32^{+}	0.247^{+}
f20	1	0.987$^{\approx}$	0.972$^{\approx}$	1.37^{+}	1.08$^{\approx}$	**0.221**$^{-}$	1.15$^{\approx}$	0.898$^{\approx}$	0.75^{-}	1.52^{+}	5.68^{+}
f21	2.04	1.92$^{\approx}$	1.8$^{\approx}$	2.82$^{\approx}$	**0.346**$^{-}$	1.8^{-}	0.93^{-}	2.04$^{\approx}$	2.52$^{\approx}$	5.27$^{\approx}$	27.5^{+}
f22	8.74	8.74$^{\approx}$	8.74$^{\approx}$	8.74$^{\approx}$	**1.32**$^{-}$	7.27$^{\approx}$	1.91^{-}	2.59$^{\approx}$	8.65$^{\approx}$	7.4$^{\approx}$	36.2^{+}
f23	**0.0735**	0.0948$^{\approx}$	0.0802$^{\approx}$	1.72^{+}	0.106$^{\approx}$	0.569^{+}	1.61^{+}	1.9^{+}	1.51^{+}	1.82^{+}	1.46^{+}
f24	62.5	58$^{\approx}$	62.1$^{\approx}$	89.9^{+}	36.8^{-}	32.6^{-}	**23.5**$^{-}$	102^{+}	109^{+}	137^{+}	182^{+}

Table 13: Median final error over 30 runs on FIR D=31 (FIR: objective value). Errors at or below 10^{-8} are shown as 0; lowest median in bold. Superscripts compare with EIG-MPHBS (paired Wilcoxon, Holm within benchmark and comparator): $^{+}$ EIG-MPHBS significantly better, $^{-}$ significantly worse, $^{\approx}$ no significant difference.

	EIG-MPHBS	MPHBS-R	MPHBS	HBA	MPA	L-SHADE	CMA-ES	DE	PSO	GWO	WOA
Case 1	4.64e-3	4.98e-3^{+}	5.01e-3^{+}	0.0113^{+}	9.09e-3^{+}	4.56e-3^{-}	**4.56e-3**$^{-}$	0.0306^{+}	0.0982^{+}	0.165^{+}	0.62^{+}
Case 2	0.0207	0.0209$^{\approx}$	0.0209$^{\approx}$	0.0351^{+}	0.0302^{+}	0.0188^{-}	**0.017**$^{-}$	0.0662^{+}	0.144^{+}	0.181^{+}	0.705^{+}
Case 3	4.61e-3	4.90e-3^{+}	5.19e-3^{+}	0.0118^{+}	9.29e-3^{+}	4.56e-3^{-}	**4.56e-3**$^{-}$	0.0321^{+}	0.0685^{+}	0.185^{+}	0.925^{+}
Case 4	0.0199	0.0213$^{\approx}$	0.0213^{+}	0.0344^{+}	0.0279^{+}	0.017^{-}	**0.017**$^{-}$	0.0616^{+}	0.141^{+}	0.166^{+}	0.96^{+}
Case 5	1.40e-3	1.42e-3$^{\approx}$	1.73e-3^{+}	5.62e-3^{+}	4.31e-3^{+}	9.68e-4^{-}	**8.88e-4**$^{-}$	0.0671^{+}	1.87^{+}	0.0837^{+}	0.943^{+}
Case 6	6.17e-3	6.21e-3$^{\approx}$	5.87e-3$^{\approx}$	9.94e-3^{+}	9.74e-3^{+}	4.70e-3^{-}	**4.70e-3**$^{-}$	0.069^{+}	0.226^{+}	0.0889^{+}	0.948^{+}
Case 7	1.37e-3	1.36e-3$^{\approx}$	1.30e-3$^{\approx}$	5.21e-3^{+}	7.21e-3^{+}	8.15e-4^{-}	**6.12e-4**$^{-}$	0.0358^{+}	2.69^{+}	0.0818^{+}	0.533^{+}
Case 8	3.23e-3	3.81e-3$^{\approx}$	3.89e-3^{+}	0.0119^{+}	8.38e-3^{+}	1.81e-3^{-}	**1.35e-3**$^{-}$	0.0117^{+}	2.58^{+}	0.096^{+}	0.48^{+}

Table 14: Mean (standard deviation) of the final error over 30 runs on CEC2022 D=10 (FIR: objective value); values at or below 10^{-8} shown as 0; lowest mean in bold.

	EIG-MPHBS	MPHBS-R	MPHBS	HBA	MPA	L-SHADE	CMA-ES	DE	PSO	GWO	WOA
F1	**0 (0)**	**0 (0)**	**0 (0)**	**0 (0)**	**0 (0)**	**0 (0)**	**0 (0)**	6.7 (17.2)	**0 (0)**	862 (1.32e+03)	8.49e+03 (1.66e+03)
F2	1.99 (3.19)	1.52 (2.63)	3.24 (3.4)	5.03 (3.13)	**0.532 (1.38)**	4.32 (1.25)	3.72 (1.01)	6.88 (2.83)	18.6 (18.5)	19.6 (19.9)	107 (90.4)
F3	1.37e-6 (2.15e-6)	2.48e-6 (7.00e-6)	1.38e-6 (3.50e-6)	0.151 (0.556)	9.79e-5 (1.95e-4)	**0 (0)**	**0 (0)**	1.29e-5 (6.80e-5)	0.448 (0.837)	0.601 (1.26)	35.9 (11)
F4	8.32 (3.24)	9.95 (4.06)	9.45 (4.4)	17.1 (7.17)	7.5 (3)	2.59 (0.964)	**0.895 (0.841)**	10.9 (5.63)	10.8 (4.57)	13.8 (7.45)	40.7 (11.7)
F5	**0 (0)**	**0 (0)**	0.0151 (0.0829)	1.36 (2.29)	2.98e-3 (0.0163)	**0 (0)**	**0 (0)**	2.99e-3 (0.0163)	0.0685 (0.148)	14.2 (35.6)	439 (152)
F6	0.198 (0.138)	0.208 (0.127)	0.203 (0.109)	2.35e+03 (2.23e+03)	**0.191 (0.142)**	0.323 (0.164)	0.636 (0.716)	9.79 (12.1)	3.18e+03 (2.31e+03)	3.22e+03 (2.48e+03)	5.08e+03 (1.11e4)
F7	2.06 (5.06)	1.06 (3.61)	2.73 (5.94)	21.3 (4.36)	0.825 (1.23)	**2.58e-4 (9.45e-4)**	7.78 (9.05)	5.82 (9.09)	19.6 (25.2)	26.1 (8.84)	92.4 (31.1)
F8	0.365 (0.356)	1.63 (5.05)	1.65 (5)	27.9 (30.7)	**0.322 (0.437)**	1.61 (2.97)	11.1 (9.73)	7.77 (9.9)	28.2 (38.5)	23.9 (4.92)	32.1 (8.71)
F9	**229 (0)**	**229 (0)**	**229 (0)**	246 (44.4)	**229 (0)**	**229 (0)**	**229 (0)**	231 (6.86)	237 (21.9)	252 (36.2)	375 (61.1)
F10	111 (32.1)	118 (40.2)	107 (26.9)	164 (61)	100 (0.0642)	100 (0.0258)	**80.7 (36.3)**	134 (57.2)	177 (52.5)	168 (54.9)	355 (335)
F11	**0 (0)**	**0 (0)**	**0 (0)**	76.7 (157)	**0 (0)**	**0 (0)**	**0 (0)**	59.8 (94.2)	180 (126)	140 (155)	480 (370)
F12	160 (1.24)	**160 (1.21)**	160 (1.47)	202 (43.5)	160 (1.48)	161 (0.978)	163 (0.386)	164 (1.4)	178 (20.3)	166 (7.93)	250 (71.7)

Table 15: Mean (standard deviation) of the final error over 30 runs on CEC2022 D=20 (FIR: objective value); values at or below 10^{-8} shown as 0; lowest mean in bold.

	EIG-MPHBS	MPHBS-R	MPHBS	HBA	MPA	L-SHADE	CMA-ES	DE	PSO	GWO	WOA
F1	**0 (0)**	2.52e-7 (1.84e-7)	9.48e-5 (5.17e-4)	1.89e-6 (5.74e-6)	1.85e-4 (3.00e-4)	**0 (0)**	**0 (0)**	93.1 (189)	2.46e+03 (6.27e+03)	5.36e+03 (4.07e+03)	3.56e4 (1.23e4)
F2	**30.1 (23.1)**	38.8 (19.3)	33.7 (21.8)	39.6 (22.4)	34.7 (22.9)	48.9 (0.765)	41.8 (14.7)	45 (15.4)	62.4 (27.4)	103 (46.8)	708 (294)
F3	3.37e-3 (0.0183)	1.10e-5 (1.43e-5)	5.54e-4 (2.90e-3)	2.31 (3.74)	0.0223 (0.0727)	1.23e-8 (1.27e-8)	**0 (0)**	3.69e-3 (0.0104)	1.46 (2.32)	2.74 (1.98)	71.6 (15.4)
F4	37.4 (13.2)	36.3 (8.28)	34.4 (11.1)	54 (18.6)	39 (11.3)	5.65 (1.05)	**3.54 (1.29)**	24.6 (12.5)	38.3 (11.5)	50.3 (25.1)	142 (24.8)
F5	22.1 (29.3)	20.1 (20.4)	41.7 (53.3)	247 (250)	20.6 (25.1)	**0 (0)**	**0 (0)**	1.73 (3.16)	31.6 (62.8)	313 (279)	2.57e+03 (799)
F6	15.5 (16.1)	23.5 (16.1)	25.4 (15.5)	6.94e+03 (8.38e+03)	11.3 (8.42)	**0.955 (0.877)**	57.8 (36.1)	5.83e+03 (6.32e+03)	1.14e6 (5.07e6)	1.94e6 (5.02e6)	1.43e8 (1.68e8)
F7	31.9 (6.36)	31.5 (7.7)	29.9 (7.75)	74.1 (37.7)	25.2 (6.09)	**15.6 (6.88)**	21.9 (1.32)	27.7 (14.1)	42.5 (35.7)	60.6 (28.4)	209 (59.9)
F8	21.1 (0.436)	**20.4 (3.43)**	21.2 (0.409)	52.1 (54.6)	20.4 (2.98)	20.6 (1.12)	21 (0.503)	23.2 (4.26)	76.4 (56.3)	44.3 (40.5)	66.3 (43.3)
F9	181 (0)	181 (0)	181 (0)	181 (0.0541)	181 (2.46e-7)	**181 (0)**	**181 (0)**	181 (0.949)	234 (48.6)	209 (21.8)	482 (169)
F10	88.2 (31.7)	85.1 (50.6)	**62.5 (47.4)**	1.35e+03 (1.43e+03)	101 (0.0918)	100 (0.0325)	92.7 (47.6)	280 (264)	448 (377)	833 (629)	2.76e+03 (1.43e+03)
F11	333 (47.9)	303 (92.8)	313 (73)	313 (97.3)	340 (122)	303 (18.3)	**250 (114)**	349 (87.6)	1.05e+03 (697)	718 (219)	3.16e+03 (1.14e+03)
F12	244 (6.05)	241 (6.83)	243 (6.52)	363 (74.7)	240 (3.44)	**235 (2.2)**	238 (4.96)	254 (21.5)	301 (46.1)	269 (28.1)	544 (165)

Table 16: Mean (standard deviation) of the final error over 30 runs on BBOB D=20 (FIR: objective value); values at or below 10^{-8} shown as 0; lowest mean in bold.

	EIG-MPHBS	MPHBS-R	MPHBS	HBA	MPA	L-SHADE	CMA-ES	DE	PSO	GWO	WOA
f1	**0 (0)**	**0 (0)**	**0 (0)**	**0 (0)**	1.56e-8 (1.69e-8)	**0 (0)**	**0 (0)**	6.74e-4 (3.34e-3)	1.55 (1.86)	0.915 (1.08)	16.7 (6.84)
f2	**0 (0)**	**0 (0)**	**0 (0)**	**0 (0)**	8.06e-4 (9.59e-4)	**0 (0)**	**0 (0)**	9.11 (21.4)	3.36e4 (8.74e4)	2.48e4 (3.15e4)	3.20e5 (2.53e5)
f3	2.06 (1.59)	1.72 (1.65)	1.26 (1.57)	36 (13)	6.24 (3.25)	**0 (0)**	9.45 (3.01)	13 (3.71)	25.4 (16.9)	47.3 (17.2)	352 (104)
f4	5.87 (2.37)	4.34 (1.66)	4.41 (1.91)	50 (18.9)	11.6 (3.72)	**0 (0)**	16.3 (2.51)	24 (8.95)	21.1 (12.8)	73.6 (26.9)	363 (73.2)
f5	**0 (0)**	**0 (0)**	**0 (0)**	93.5 (33)	**0 (0)**	**0 (0)**	**0 (0)**	**0 (0)**	29.2 (23.7)	12.2 (8.41)	104 (44.7)
f6	6.80e-8 (7.79e-8)	4.66e-6 (8.61e-6)	6.98e-6 (9.06e-6)	4.35e-3 (0.019)	4.06e-4 (8.42e-4)	**0 (0)**	**0 (0)**	0.104 (0.568)	137 (94.1)	44.4 (27.2)	2.12e4 (3.69e4)
f7	3.8 (2.57)	4.19 (2.58)	3.59 (2.51)	8.57 (4.63)	2.45 (1.68)	**0 (0)**	**0 (0)**	3.27 (2.18)	16.9 (17.9)	15.6 (8.59)	190 (85.9)
f8	0.133 (0.728)	0.139 (0.727)	0.507 (1.2)	2.74 (2.31)	10.9 (11.9)	**0 (0)**	**0 (0)**	23.4 (20.5)	252 (614)	141 (130)	3.45e+03 (2.32e+03)
f9	0.284 (1.02)	1.33 (1.58)	1.96 (1.68)	5.73 (2.97)	4.41 (3.24)	**0 (0)**	**0 (0)**	24.5 (21.8)	25.7 (23.7)	30.9 (17)	102 (14.6)
f10	16.1 (16.5)	139 (189)	139 (99.5)	883 (468)	108 (124)	**0 (0)**	**0 (0)**	656 (403)	8.36e+03 (6.55e+03)	1.31e4 (8.09e+03)	2.65e5 (1.71e5)
f11	1.53e-8 (1.79e-8)	0.012 (0.0356)	0.0248 (0.0517)	18.4 (92.8)	2.23e-3 (3.85e-3)	**0 (0)**	**0 (0)**	3.28 (2.31)	5.5 (9.37)	40 (13.7)	118 (52.5)
f12	10.6 (18.8)	5.07 (5.78)	10.3 (13)	8.14 (12)	7.84 (13.7)	1.13e-8 (0)	**0 (0)**	5.50e4 (3.01e5)	1.65e6 (3.25e6)	6.79e5 (9.65e5)	2.89e7 (1.40e7)
f13	2.04 (2.83)	6.18 (8.01)	5.14 (6.32)	10 (12.4)	4.13 (5.76)	**0 (0)**	**0 (0)**	6.39 (9.11)	236 (209)	192 (130)	872 (180)
f14	1.09e-5 (5.72e-6)	8.47e-5 (3.48e-5)	8.80e-5 (4.25e-5)	7.99e-5 (2.09e-5)	1.40e-4 (1.16e-4)	**0 (0)**	**0 (0)**	1.53e-4 (2.24e-4)	1.71 (1.63)	0.656 (0.895)	9.13 (3.35)
f15	57.6 (19.9)	55.6 (19.7)	56.8 (18.1)	76.2 (26.7)	40.2 (14.3)	3.93 (1.15)	**3.85 (1.56)**	22.7 (13.5)	58.9 (25.3)	77.5 (34.2)	370 (83)
f16	2.06 (1.22)	2.13 (1.02)	1.95 (0.845)	6.48 (3.84)	1.44 (0.781)	3.95 (0.992)	**0.0569 (0.0602)**	10.3 (5.41)	3.33 (1.52)	1.51 (0.744)	16.5 (5.24)
f17	0.619 (0.514)	0.687 (0.478)	0.75 (0.78)	1.71 (0.97)	0.307 (0.194)	**2.03e-8 (5.65e-8)**	2.04e-8 (5.68e-8)	0.0863 (0.112)	0.863 (0.581)	0.951 (0.612)	7.67 (1.74)
f18	1.55 (1.21)	1.45 (1.15)	1.25 (0.829)	5.72 (3.07)	1.38 (1.17)	**1.20e-4 (5.21e-4)**	3.81e-3 (9.04e-3)	0.441 (0.579)	2.2 (1.43)	2.69 (1.14)	28.8 (7.15)
f19	**0.0157 (0)**	0.067 (0.102)	0.0605 (0.103)	0.337 (0.901)	0.199 (0.127)	1.07 (0.297)	0.654 (0.378)	3.86 (0.4)	3.59 (0.442)	3.25 (0.379)	1.37 (1.57)
f20	1.02 (0.164)	0.97 (0.202)	0.933 (0.179)	1.35 (0.231)	1.06 (0.175)	**0.215 (0.0822)**	1.11 (0.256)	0.908 (0.208)	38.3 (206)	1.51 (0.2)	202 (393)
f21	5.03 (6.4)	4.71 (5.55)	3.5 (5.62)	6.05 (7.5)	**0.82 (0.902)**	2.71 (4.23)	1.09 (0.895)	4.1 (6.23)	6.41 (8.1)	7.38 (7.68)	31.2 (16.9)
f22	10.8 (9.8)	10.8 (10.3)	9.77 (9.94)	11.9 (13.6)	**2.6 (2.97)**	7.22 (6.73)	2.67 (2.81)	7.22 (7.54)	11.5 (11.6)	8.21 (8.02)	36.9 (16.7)
f23	**0.0833 (0.0446)**	0.116 (0.0754)	0.0957 (0.0467)	1.7 (0.382)	0.121 (0.0792)	0.563 (0.138)	1.3 (0.817)	1.84 (0.311)	1.47 (0.247)	1.79 (0.253)	1.55 (0.55)
f24	63.1 (14.7)	59.8 (14.3)	62.7 (16.1)	89.3 (23.9)	36.5 (10.7)	32.6 (1.99)	**20 (10.2)**	101 (15.3)	108 (17.8)	132 (18.7)	179 (19)

Table 17: Mean (standard deviation) of the final error over 30 runs on FIR D=31 (FIR: objective value); values at or below 10^{-8} shown as 0; lowest mean in bold.

	EIG-MPHBS	MPHBS-R	MPHBS	HBA	MPA	L-SHADE	CMA-ES	DE	PSO	GWO	WOA
Case 1	4.88e-3 (7.32e-4)	5.41e-3 (9.01e-4)	5.35e-3 (9.72e-4)	0.0194 (0.0187)	0.0114 (6.98e-3)	4.56e-3 (1.49e-6)	**4.56e-3 (0)**	0.0704 (0.149)	0.969 (3.31)	0.209 (0.128)	0.629 (0.0326)
Case 2	0.0202 (2.67e-3)	0.0209 (2.41e-3)	0.0203 (2.05e-3)	0.0514 (0.0391)	0.0302 (5.96e-3)	0.0188 (1.79e-3)	**0.017 (0)**	0.0794 (0.0588)	2.18 (4.62)	0.215 (0.114)	0.687 (0.0721)
Case 3	4.68e-3 (1.81e-4)	5.15e-3 (9.01e-4)	5.45e-3 (1.02e-3)	0.0174 (0.0143)	0.0104 (4.22e-3)	4.56e-3 (1.72e-6)	**4.56e-3 (0)**	0.0759 (0.16)	3.13 (6.4)	0.199 (0.0942)	0.916 (0.0388)
Case 4	0.0194 (2.34e-3)	0.0212 (2.43e-3)	0.0209 (2.06e-3)	0.0415 (0.0189)	0.0304 (8.10e-3)	0.0184 (1.75e-3)	**0.017 (0)**	0.106 (0.104)	2.14 (6.44)	0.194 (0.107)	0.952 (0.0236)
Case 5	1.49e-3 (4.02e-4)	1.71e-3 (7.36e-4)	1.92e-3 (6.23e-4)	9.42e-3 (8.55e-3)	5.57e-3 (3.64e-3)	9.93e-4 (1.31e-4)	**8.88e-4 (0)**	0.0779 (0.0495)	2.62 (4.33)	0.0941 (0.0708)	0.928 (0.0464)
Case 6	6.46e-3 (1.46e-3)	6.65e-3 (1.64e-3)	6.37e-3 (1.65e-3)	0.0137 (9.09e-3)	0.0106 (3.74e-3)	4.72e-3 (4.29e-5)	**4.70e-3 (0)**	0.142 (0.257)	2.83 (5.83)	0.146 (0.135)	0.941 (0.0377)
Case 7	1.49e-3 (5.36e-4)	1.45e-3 (5.08e-4)	1.61e-3 (9.05e-4)	0.012 (0.0172)	7.85e-3 (4.96e-3)	7.90e-4 (1.08e-4)	**6.42e-4 (1.20e-4)**	0.0434 (0.0297)	2.58 (1.78)	0.097 (0.0617)	0.502 (0.178)
Case 8	3.35e-3 (5.35e-4)	4.19e-3 (1.67e-3)	4.92e-3 (2.55e-3)	0.0188 (0.0167)	8.41e-3 (3.35e-3)	2.05e-3 (5.01e-4)	**1.43e-3 (9.69e-5)**	0.0152 (8.75e-3)	3.2 (2.77)	0.111 (0.045)	0.496 (0.147)

8. Discussion

What the eigen frame achieves. The exchange step of HBA–MPA hybrids is where the two populations combine their information, and it was written for separable problems. Moving it into the eigen frame of the pooled populations makes it invariant to rotations at no extra evaluation cost. The mechanism analysis shows why this matters: on coupled problems the axis-frame exchange stops producing acceptable candidates early in the run, and the eigen frame restores it. The results follow: EIG-MPHBS is never significantly worse than MPHBS on any of 56 instances and is better on 14, with the largest gains on ill-conditioned rotated functions, on CEC2022 at $D = 20$ and on the FIR application. The change is local, a few lines in the mediator, and should transfer to other hybrids that assemble solutions coordinate by coordinate.

What it does not achieve. The frame helps unimodal or moderately multimodal problems with coupled variables, where the population covariance reflects the basin; it does not help highly multimodal rotated functions such as f15, and it costs a little on separable multimodal ones such as f3 and f4, where half of the exchanges no longer follow the separable axes. Phase 1 remains coordinate-wise, which is why the gap to CMA-ES and L-SHADE on rotated functions persists. The comparison with EA4eig confirms that this is an improvement of a hybrid within its class, not a new state of the art.

Practical guidance. For practitioners who use HBA–MPA hybrids, for example because they are already part of an application pipeline, EIG-MPHBS is a safe replacement for MPHBS: same budget, similar run time, never worse in our tests and often better. When the choice of algorithm is open, L-SHADE and CMA-ES remain the stronger default on problems of this size, and the classical metaheuristics PSO, GWO and WOA are clearly weaker than all three hybrids.

Threats to validity. (i) The confirmatory effect is small and sensitive to the handling of tied instances; the stronger effects come from instances that include the pilot instances. (ii) The value $q = 0.5$ was chosen from a pilot with two values; other values or an adaptive q may perform differently. (iii) The six classical algorithms were added after the main analysis and are reported descriptively; they use standard settings, not tuned ones. (iv) The EA4eig comparison is unpaired and across implementations. (v) Run times were measured on a shared laptop. (vi) All problems have $D \leq 31$; for larger D the $O(D^3)$ decomposition and the reliability of the covariance estimate from $2N$ points should be checked.

9. Conclusion

We identified the coordinate dependence of the dimension-wise exchange in the HBA–MPA hybrid MPHBS as a structural weakness on rotated problems and proposed EIG-MPHBS, which performs the exchange in the eigenvector frame of the pooled populations in half of the iterations. The exchange becomes provably rotation invariant at no extra evaluation cost and a few percent of extra run time, and on coupled problems it restores the acceptance of exchanged candidates that the axis frame loses. In 18,840 runs on CEC2022, BBOB and FIR filter design, EIG-MPHBS improved on MPHBS on 14 instances and was

never worse, passed a test fixed in advance on 28 held-out instances (narrowly), and outperformed its parent algorithms and four classical metaheuristics. It ranked third of eleven, behind L-SHADE and CMA-ES, and remained behind EA4eig at the CEC2022 budgets. Future work includes an adaptive choice of the frame probability, rotation-invariant versions of the Phase-1 moves, and tests at higher dimension and on further applications.

Data and code availability

Code, the design document (`experiments/ESWA_EIGEN_CAMPAIGN.md`), the record of each of the 18,840 runs (final error, evaluations, run time, configuration and seed), the analyses and the scripts that generate every number and table in this paper are available at a public repository whose link is withheld for double-anonymized review and given on the title page. The per-run convergence and population traces used for the figures (1.6 GB) are available from the corresponding author on request. The EA4eig results are the official CEC2022 competition files, redistributed with their source in `data/external/cec2022_ea4eig/`.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

References

Adam, A.H.A., Hassan, M.H., Kamel, S., 2027. QLRUN: Integrating Q-learning and covariance matrix learning into Runge–Kutta optimization for engineering design problems. Unconventional Resources 16, 100484. doi:10.1016/j.uncres.2026.100484.

Al-Betar, M.A., Awadallah, M.A., Makhadmeh, S.N., Alyasseri, Z.A.A., Al-Naymat, G., Mirjalili, S., 2023. Marine predators algorithm: A review. Archives of Computational Methods in Engineering 30, 3405–3435. doi:10.1007/s11831-023-09912-1.

Al-Dhaifallah, M., Refaat, M.M., Alaas, Z., Abdel Aleem, S.H.E., Ali, Z.M., 2024. Enhancing hosting capacity for electric vehicles in modern power networks using improved hybrid optimization approaches with environmental sustainability considerations. Scientific Reports 14, 25607. doi:10.1038/s41598-024-76410-0.

Alba, E., Tomassini, M., 2002. Parallelism and evolutionary algorithms. IEEE Transactions on Evolutionary Computation 6, 443–462. doi:10.1109/TEVC.2002.800880.

Auger, A., Hansen, N., 2005. A restart CMA evolution strategy with increasing population size, in: 2005 IEEE Congress on Evolutionary Computation, pp. 1769–1776.

Blum, C., Puchinger, J., Raidl, G.R., Roli, A., 2011. Hybrid metaheuristics in combinatorial optimization: A survey. Applied Soft Computing 11, 4135–4151.

Bujok, P., Kolenovský, P., 2022. Eigen crossover in cooperative model of evolutionary algorithms applied to CEC 2022 single objective numerical optimisation, in: 2022 IEEE Congress on Evolutionary Computation (CEC), pp. 1–8. doi:10.1109/CEC55065.2022.9870433.

Civicioglu, P., Besdok, E., 2018. A+ evolutionary search algorithm and QR decomposition based rotation invariant crossover operator. Expert Systems with Applications 103, 49–62. doi:10.1016/j.eswa.2018.03.009.

Črepinšek, M., Liu, S.H., Mernik, M., 2013. Exploration and exploitation in evolutionary algorithms: A survey. ACM Computing Surveys 45, 35:1–35:33.

Demšar, J., 2006. Statistical comparisons of classifiers over multiple data sets. Journal of Machine Learning Research 7, 1–30.

Derrac, J., García, S., Molina, D., Herrera, F., 2011. A practical tutorial on the use of nonparametric statistical tests as a methodology for comparing evolutionary and swarm intelligence algorithms. Swarm and Evolutionary Computation 1, 3–18.

Drugan, M.M., 2019. Reinforcement learning versus evolutionary computation: A survey on hybrid algorithms. Swarm and Evolutionary Computation 44, 228–246.

Faramarzi, A., Heidarinejad, M., Mirjalili, S., Gandomi, A.H., 2020. Marine predators algorithm: A nature-inspired metaheuristic. Expert Systems with Applications 152, 113377.

Fialho, Á., Da Costa, L., Schoenauer, M., Sebag, M., 2010. Analyzing bandit-based adaptive operator selection mechanisms. Annals of Mathematics and Artificial Intelligence 60, 25–64.

Flores, E., Olivares, R., 2026. MPHBS: A novel hybrid metaheuristic with SARSA-guided information exchange. Knowledge-Based Systems 352, 117070.

Guo, S.M., Yang, C.C., 2015. Enhancing differential evolution utilizing eigenvector-based crossover operator. IEEE Transactions on Evolutionary Computation 19, 31–49.

Hamad, Q.S., Samma, H., Suandi, S.A., Mohamad-Saleh, J., 2022. Q-learning embedded sine cosine algorithm (QLESCA). Expert Systems with Applications 193, 116417. doi:10.1016/j.eswa.2021.116417.

Hansen, N., Akimoto, Y., Baudis, P., 2019. CMA-ES/pycma on Github. Zenodo. doi:10.5281/zenodo.2559634.

Hansen, N., Auger, A., Ros, R., Mersmann, O., Tušar, T., Brockhoff, D., 2021. COCO: a platform for comparing continuous optimizers in a black-box setting. Optimization Methods and Software 36, 114–144.

Hansen, N., Finck, S., Ros, R., Auger, A., 2009. Real-Parameter Black-Box Optimization Benchmarking 2009: Noiseless Functions Definitions. Technical Report RR-6829. INRIA.

Hansen, N., Ostermeier, A., 2001. Completely derandomized self-adaptation in evolution strategies. Evolutionary Computation 9, 159–195.

Hashim, F.A., Houssein, E.H., Hussain, K., Mabrouk, M.S., Al-Atabany, W., 2022. Honey badger algorithm: New metaheuristic algorithm for solving optimization problems. Mathematics and Computers in Simulation 192, 84–110.

Hassan, I.H., Abdullahi, M., Isuwa, J., Yusuf, S.A., Aliyu, I.T., 2024. A comprehensive survey of honey badger optimization algorithm and meta-analysis of its variants and applications. Franklin Open 8, 100141.

Holm, S., 1979. A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics 6, 65–70.

Karafotias, G., Hoogendoorn, M., Eiben, A.E., 2015. Parameter control in evolutionary algorithms: Trends and challenges. IEEE Transactions on Evolutionary Computation 19, 167–187.

Kennedy, J., Eberhart, R., 1995. Particle swarm optimization, in: Proceedings of ICNN'95 – International Conference on Neural Networks, pp. 1942–1948.

Kůdela, J., 2022. A critical problem in benchmarking and analysis of evolutionary computation methods. Nature Machine Intelligence 4, 1238–1245. doi:10.1038/s42256-022-00579-0.

Kumar, A., Price, K.V., Mohamed, A.W., Hadi, A.A., Suganthan, P.N., 2021. Problem Definitions and Evaluation Criteria for the CEC 2022 Special Session and Competition on Single Objective Bound Constrained Numerical Optimization. Technical Report. Nanyang Technological University.

Ma, H., Shen, S., Yu, M., Yang, Z., Fei, M., Zhou, H., 2019. Multi-population techniques in nature inspired optimization algorithms: A comprehensive survey. Swarm and Evolutionary Computation 44, 365–387. doi:10.1016/j.swevo.2018.04.011.

Mirjalili, S., Lewis, A., 2016. The whale optimization algorithm. Advances in Engineering Software 95, 51–67.

Mirjalili, S., Mirjalili, S.M., Lewis, A., 2014. Grey wolf optimizer. Advances in Engineering Software 69, 46–61.

Morales-Castañeda, B., Zaldivar, D., Cuevas, E., Fausto, F., Rodríguez, A., 2020. A better balance in metaheuristic algorithms: Does it exist? Swarm and Evolutionary Computation 54, 100671.

Salomon, R., 1996. Re-evaluating genetic algorithm performance under coordinate rotation of benchmark functions. A survey of some theoretical and practical aspects of genetic algorithms. BioSystems 39, 263–278.

Seyyedabbasi, A., Aliyev, R., Kiani, F., Gulle, M.U., Basyildiz, H., Shah, M.A., 2021. Hybrid algorithms based on combining reinforcement learning and metaheuristic methods to solve global optimization problems. Knowledge-Based Systems 223, 107044.

Shi, Y., Eberhart, R., 1998. A modified particle swarm optimizer, in: 1998 IEEE International Conference on Evolutionary Computation, pp. 69–73.

Song, Y., Wu, Y., Guo, Y., Yan, R., Suganthan, P.N., Zhang, Y., Pedrycz, W., Das, S., Mallipeddi, R., Ajani, O.S., Feng, Q., 2024. Reinforcement learning-assisted evolutionary algorithm: A survey and research opportunities. Swarm and Evolutionary Computation 86, 101517. doi:10.1016/j.swevo.2024.101517.

Squillero, G., Tonda, A., 2016. Divergence of character and premature convergence: A survey of methodologies for promoting diversity in evolutionary optimization. Information Sciences 329, 782–799. doi:10.1016/j.ins.2015.09.056.

Storn, R., Price, K., 1997. Differential evolution – a simple and efficient heuristic for global optimization over continuous spaces. Journal of Global Optimization 11, 341–359.

Talbi, E.G., 2002. A taxonomy of hybrid metaheuristics. Journal of Heuristics 8, 541–564.

Talbi, E.G., 2021. Machine learning into metaheuristics: A survey and taxonomy. ACM Computing Surveys 54, 129:1–129:32.

Tanabe, R., Fukunaga, A.S., 2014. Improving the search performance of SHADE using linear population size reduction, in: 2014 IEEE Congress on Evolutionary Computation (CEC), pp. 1658–1665.

Tang, Z., Gong, M., Wu, Y., Liu, W., Xie, Y., 2021. Regularized evolutionary multitask optimization: Learning to intertask transfer in aligned subspace. IEEE Transactions on Evolutionary Computation 25, 262–276.

Tang, Z., Gong, M., Wu, Y., Qin, A.K., Tan, K.C., 2022. A multifactorial optimization framework based on adaptive intertask coordinate system. IEEE Transactions on Cybernetics 52, 6745–6758.

Vargha, A., Delaney, H.D., 2000. A critique and improvement of the CL common language effect size statistics of McGraw and Wong. Journal of Educational and Behavioral Statistics 25, 101–132.

Wang, Y., Li, H.X., Huang, T., Li, L., 2014. Differential evolution based on covariance matrix learning and bimodal distribution parameter setting. Applied Soft Computing 18, 232–247.

Wilcoxon, F., 1945. Individual comparisons by ranking methods. Biometrics Bulletin 1, 80–83.

Wolpert, D.H., Macready, W.G., 1997. No free lunch theorems for optimization. IEEE Transactions on Evolutionary Computation 1, 67–82.